

BFS Traversal

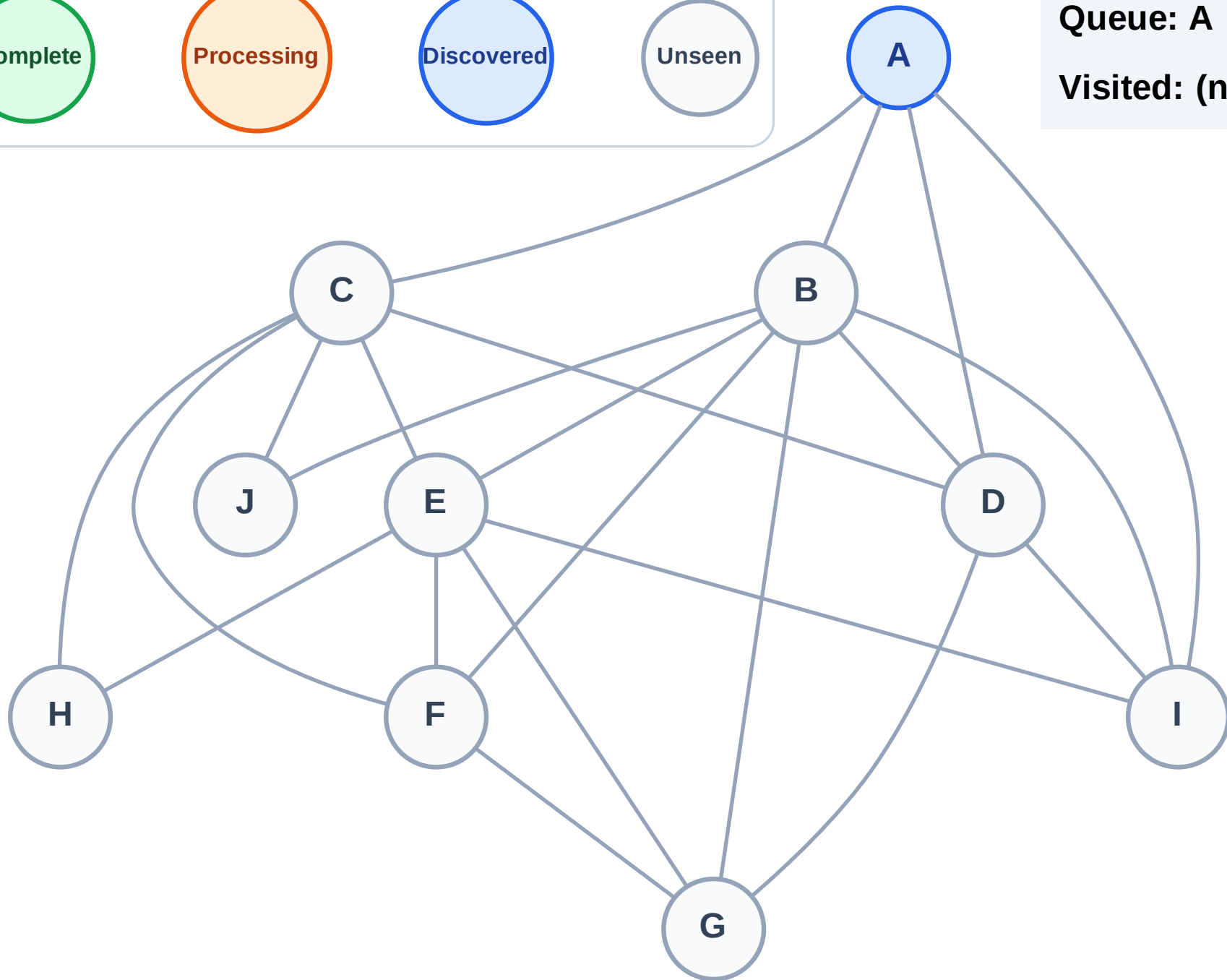
Step 1: Start at A

Legend



Queue: A

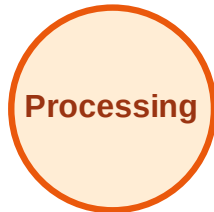
Visited: (none)



BFS Traversal

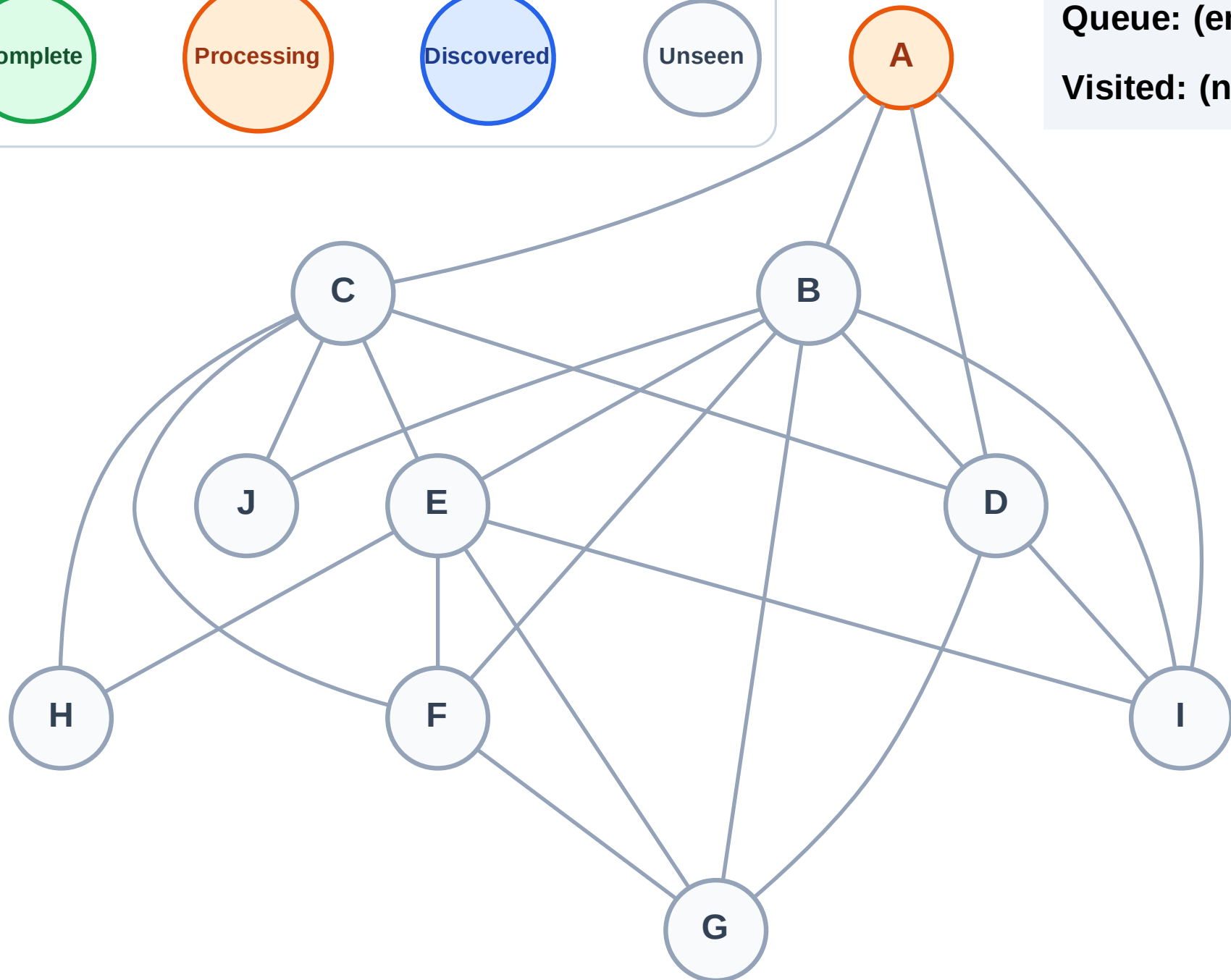
Step 2: Process node A

Legend



Queue: (empty)

Visited: (none)



BFS Traversal

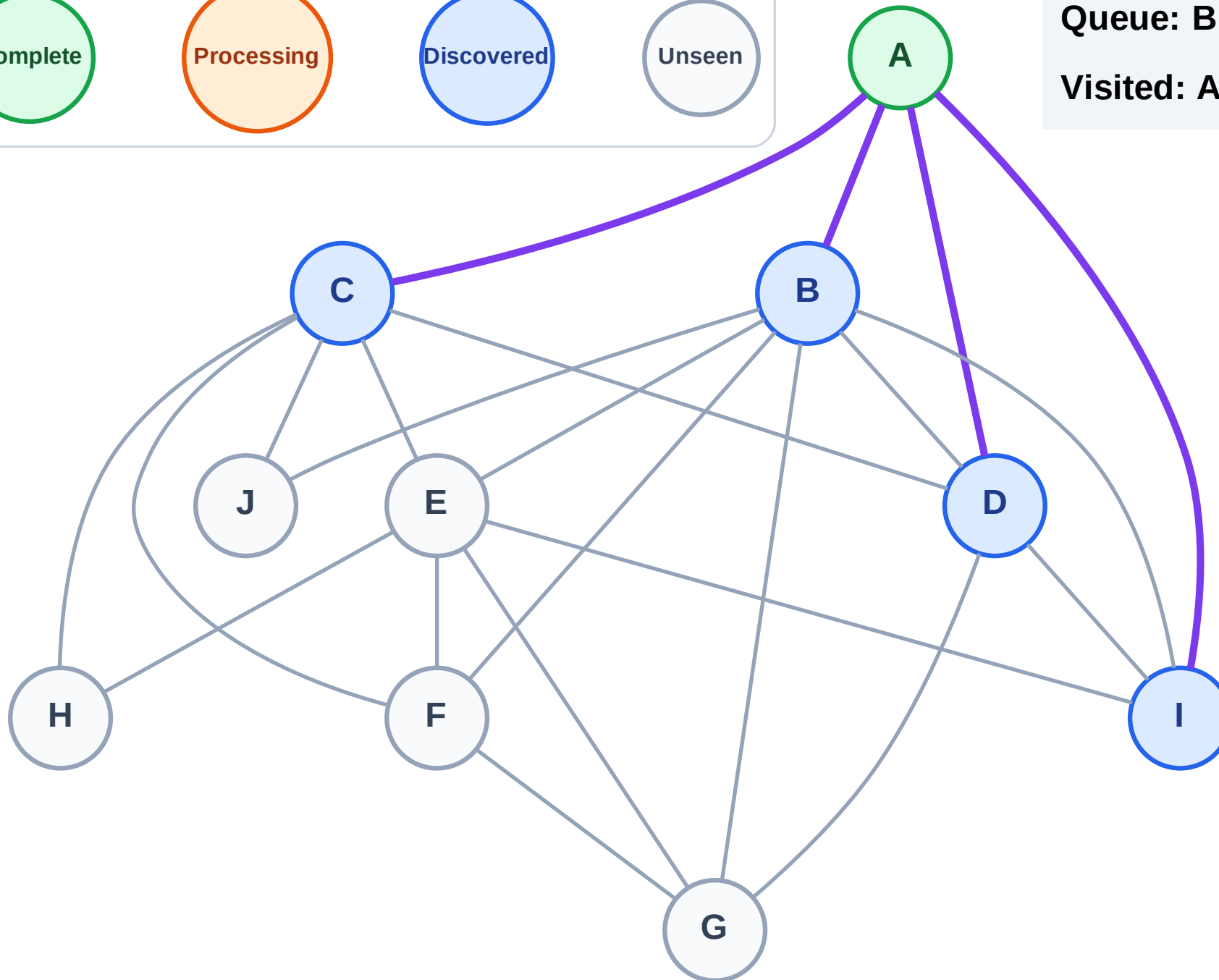
Step 3: Discover B, C, D, I from A

Legend



Queue: B → C → D → I

Visited: A



BFS Traversal

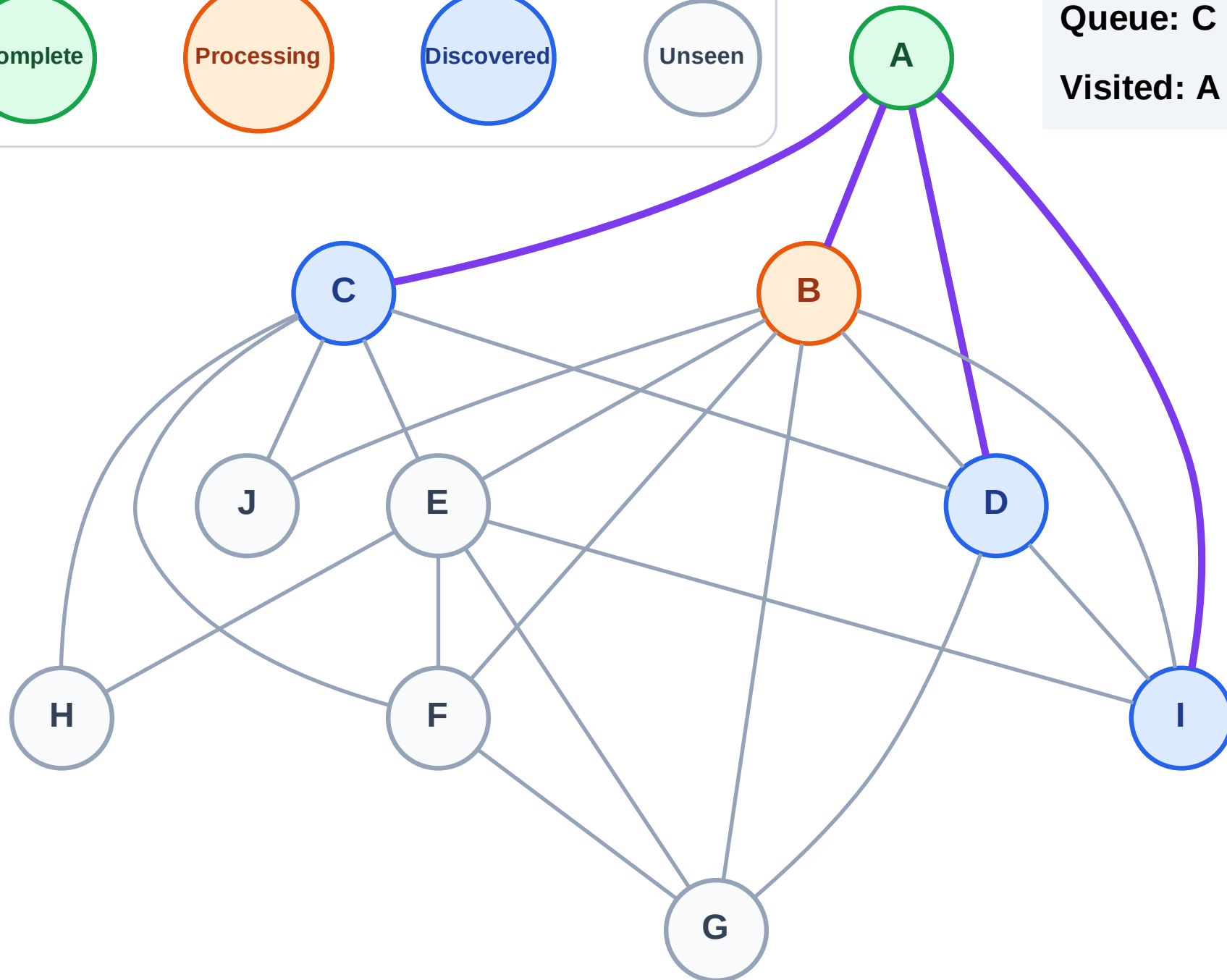
Step 4: Process node B

Legend



Queue: C → D → I

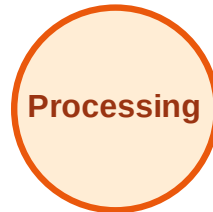
Visited: A



BFS Traversal

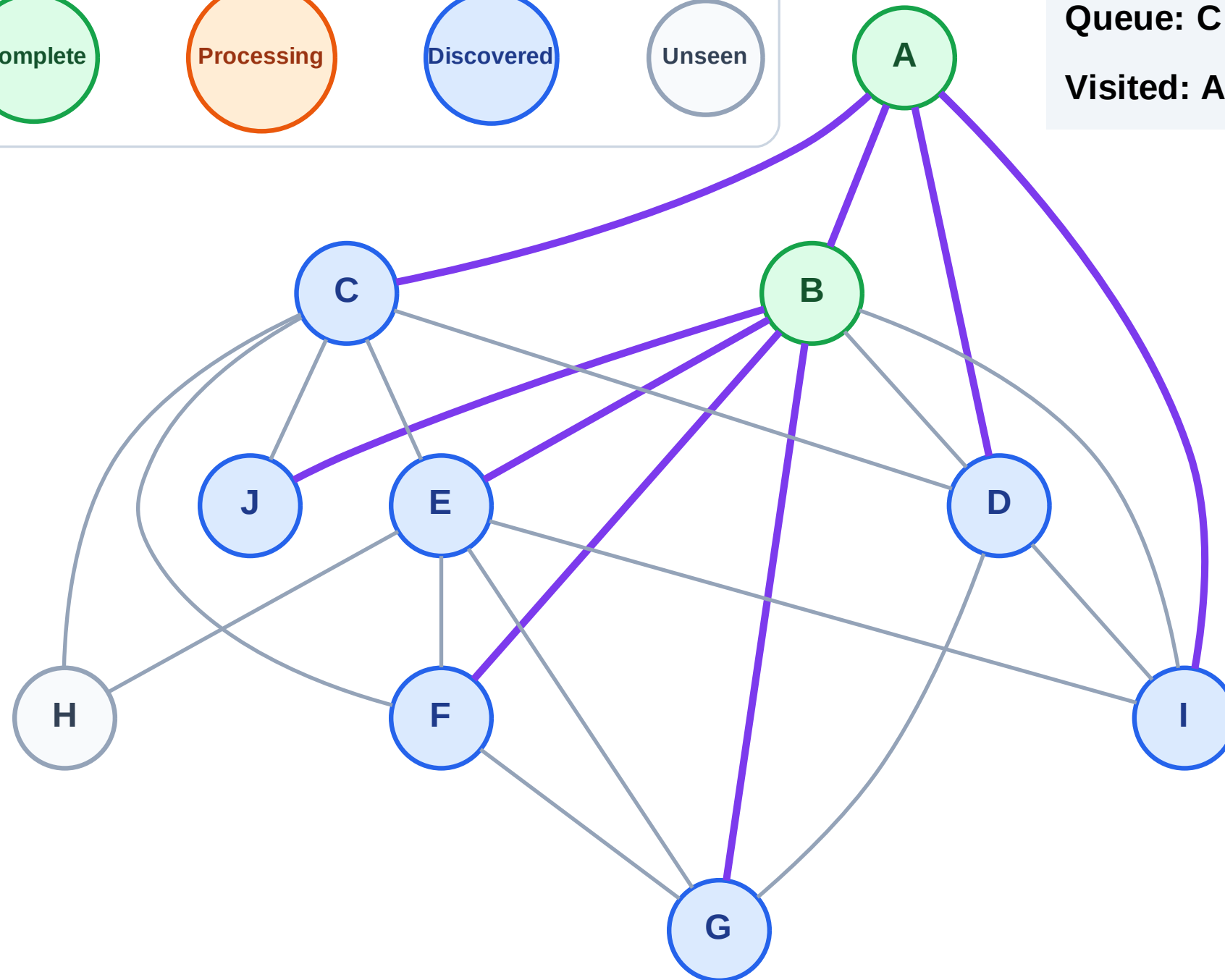
Step 5: Discover E, F, G, J from B

Legend



Queue: C → D → I → E → F → G → J

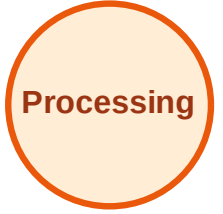
Visited: A, B



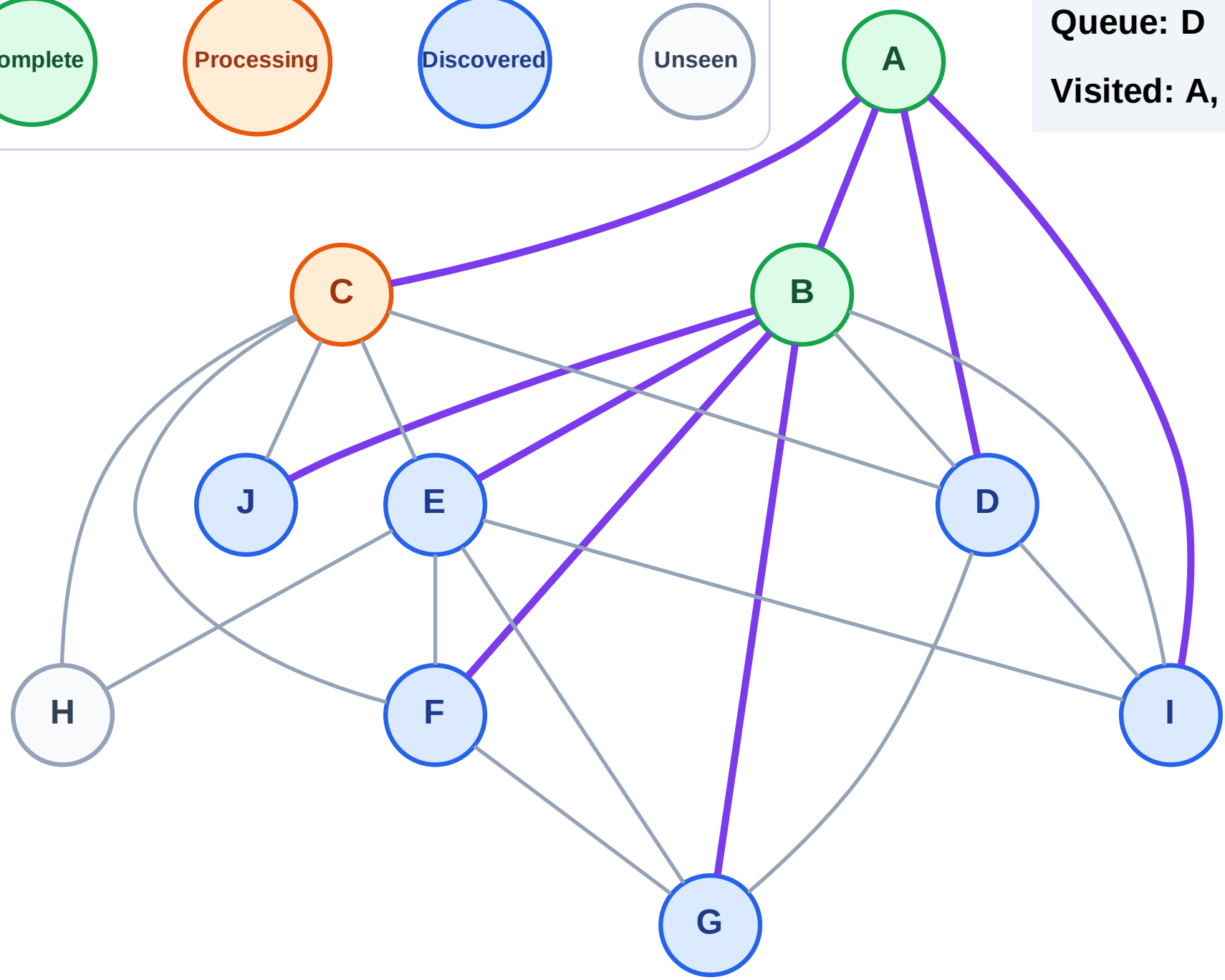
BFS Traversal

Step 6: Process node C

Legend



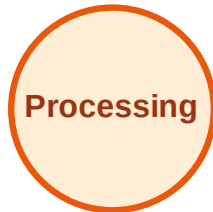
Queue: D → I → E → F → G → J
Visited: A, B



BFS Traversal

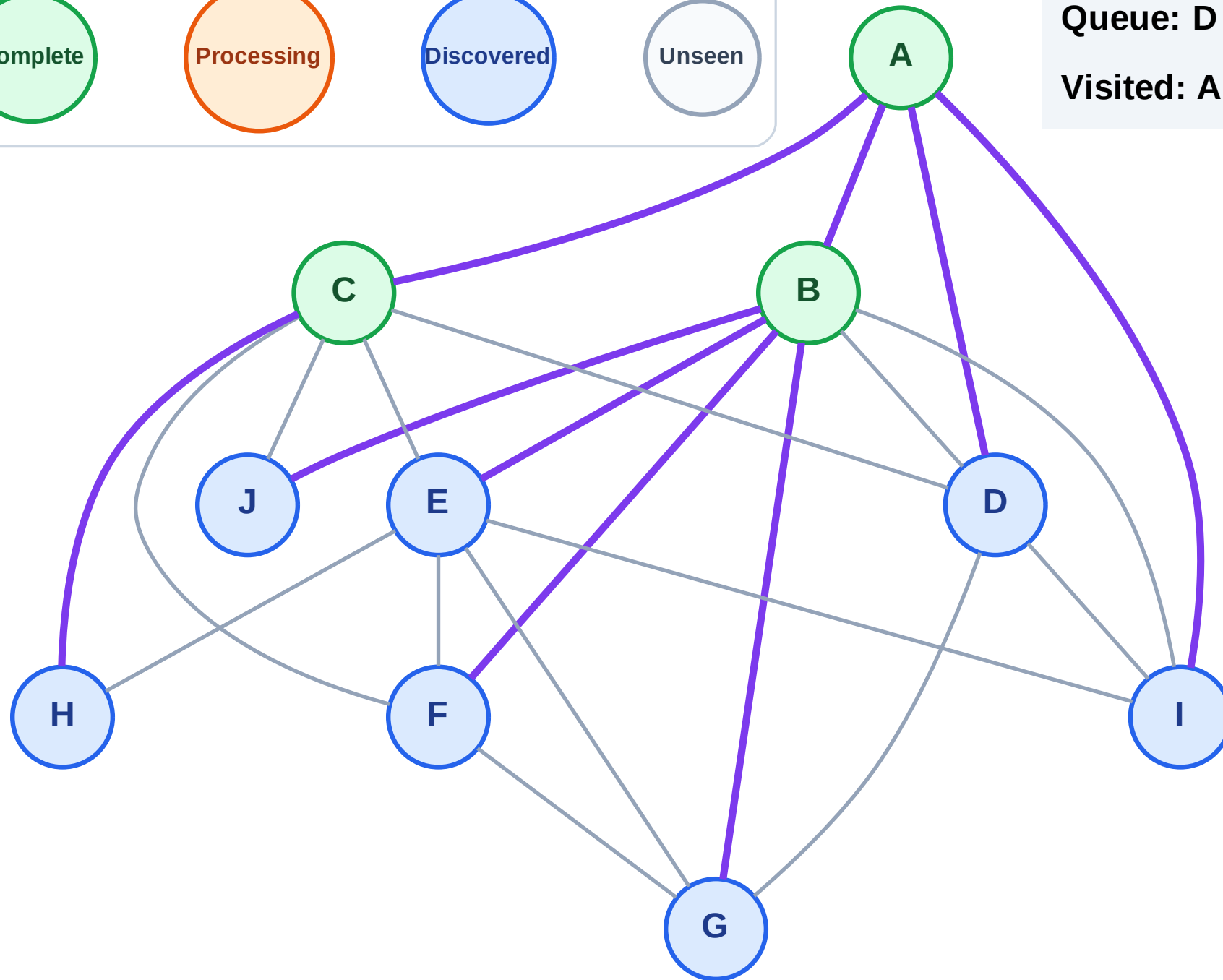
Step 7: Discover H from C

Legend



Queue: D → I → E → F → G → J → H

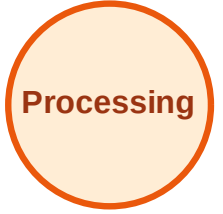
Visited: A, B, C



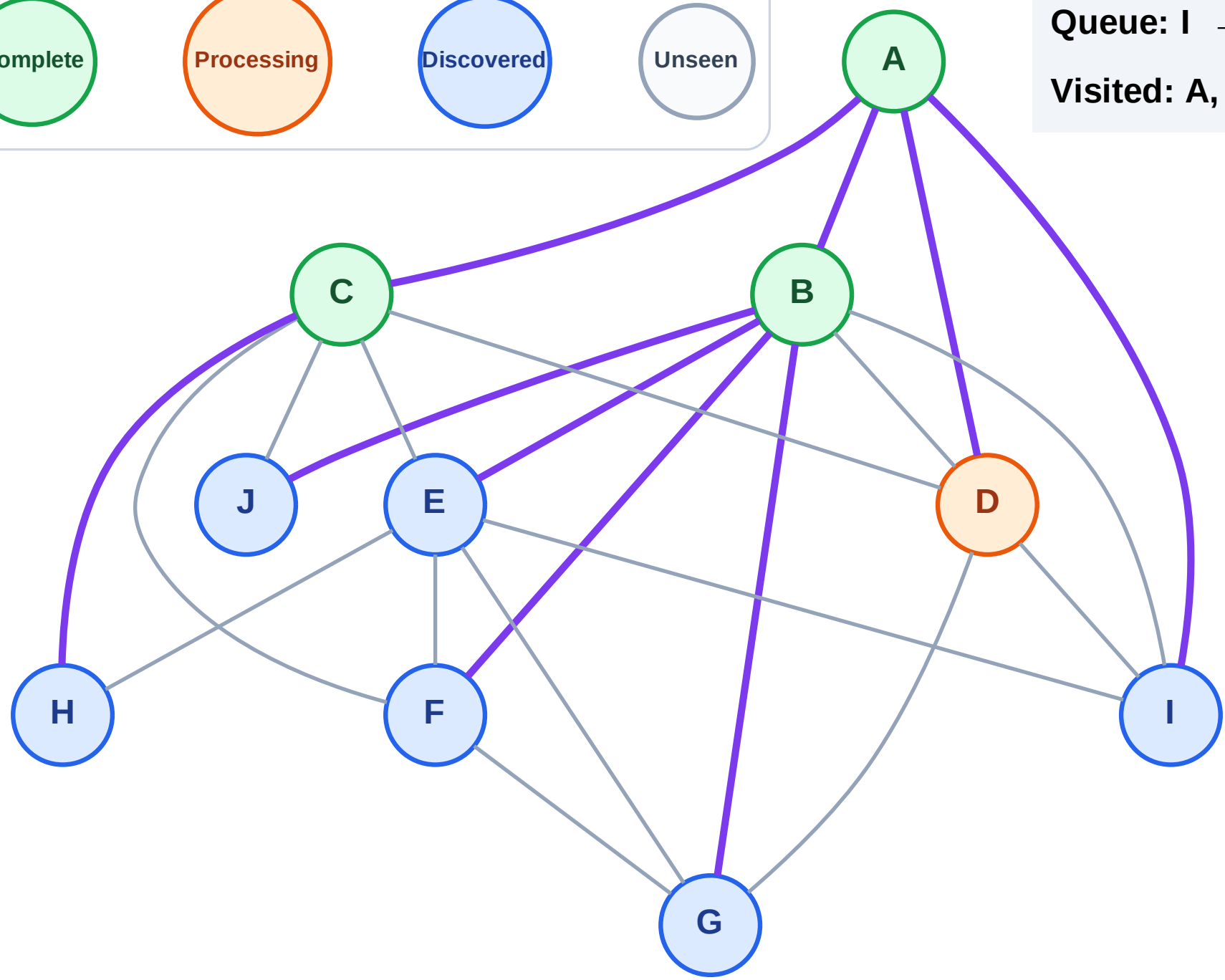
BFS Traversal

Step 8: Process node D

Legend



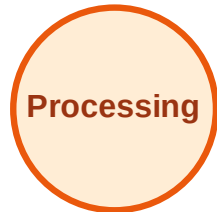
Queue: I → E → F → G → J → H
Visited: A, B, C



BFS Traversal

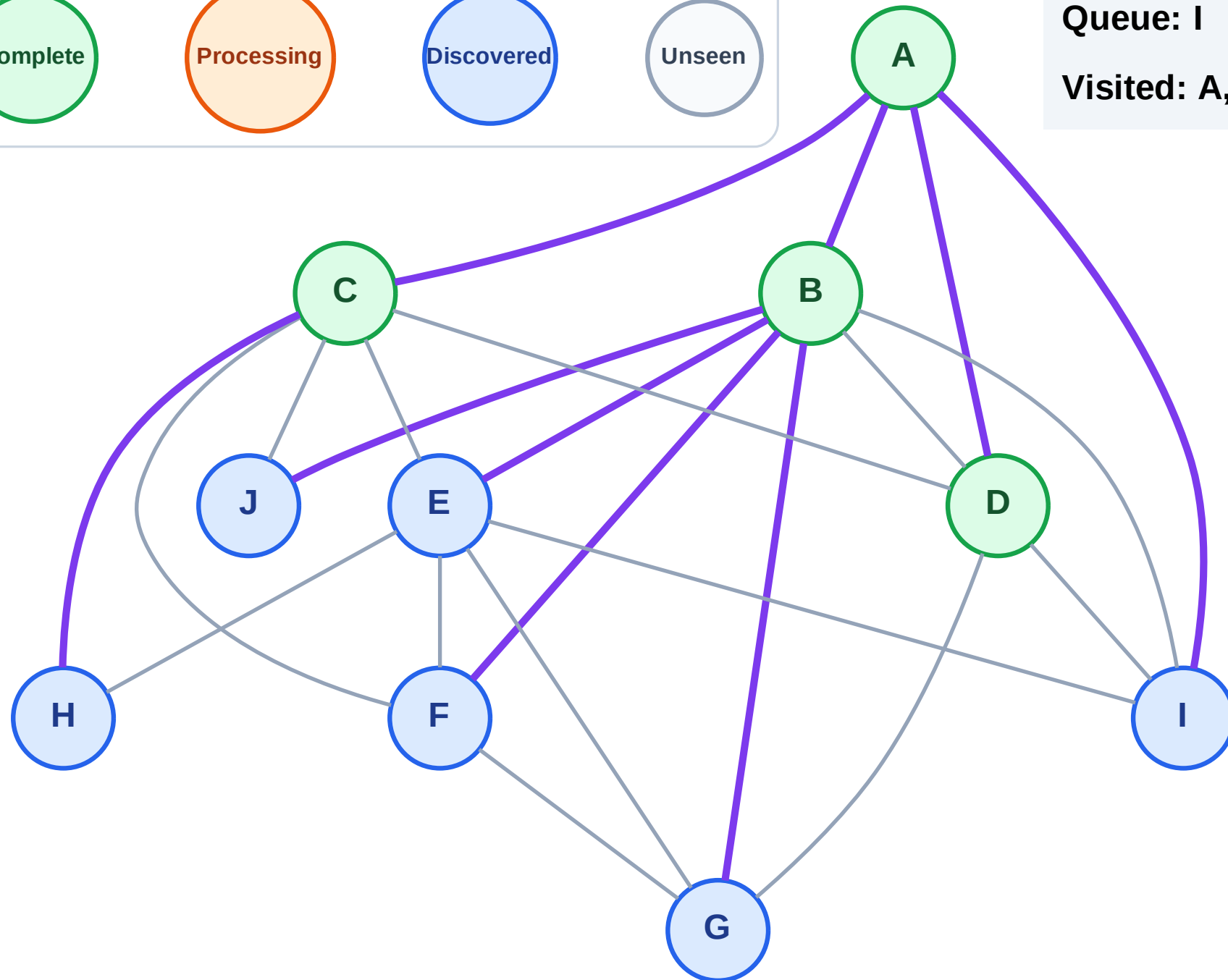
Step 9: No new neighbors from D

Legend



Queue: I → E → F → G → J → H

Visited: A, B, C, D



BFS Traversal

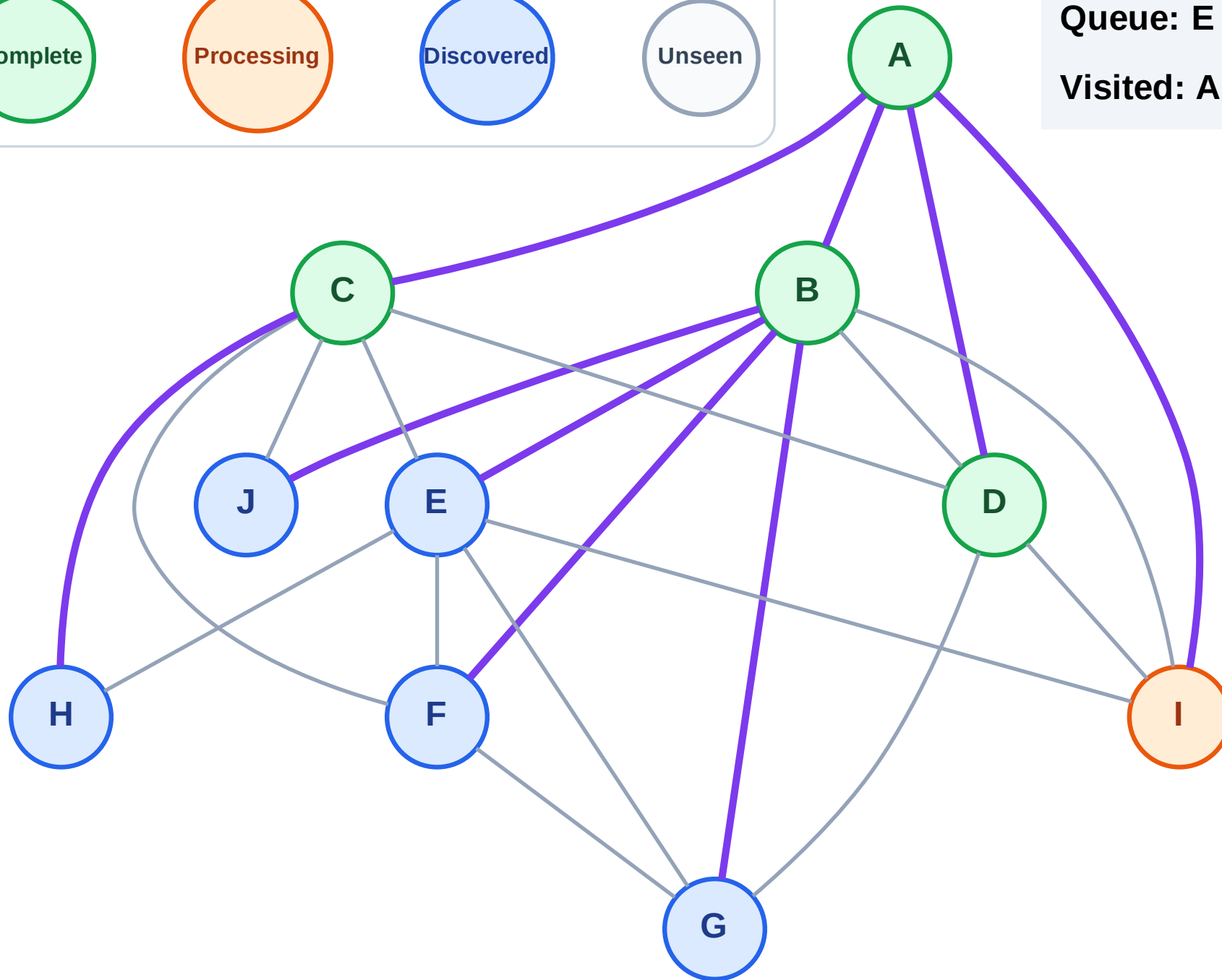
Step 10: Process node I

Legend



Queue: E → F → G → J → H

Visited: A, B, C, D



BFS Traversal

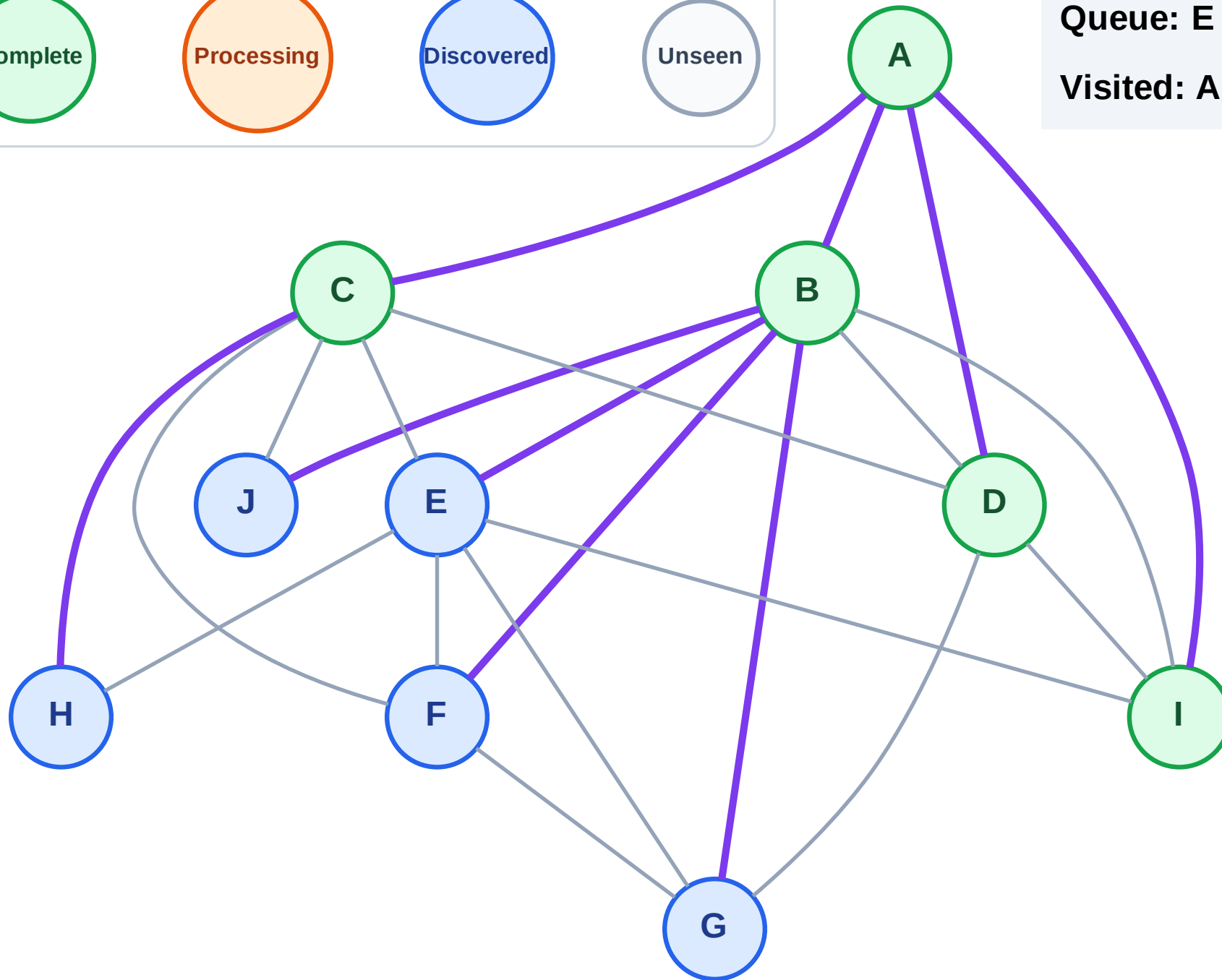
Step 11: No new neighbors from I

Legend



Queue: E → F → G → J → H

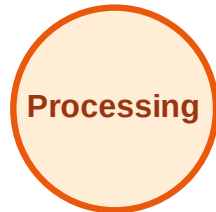
Visited: A, B, C, D, I



BFS Traversal

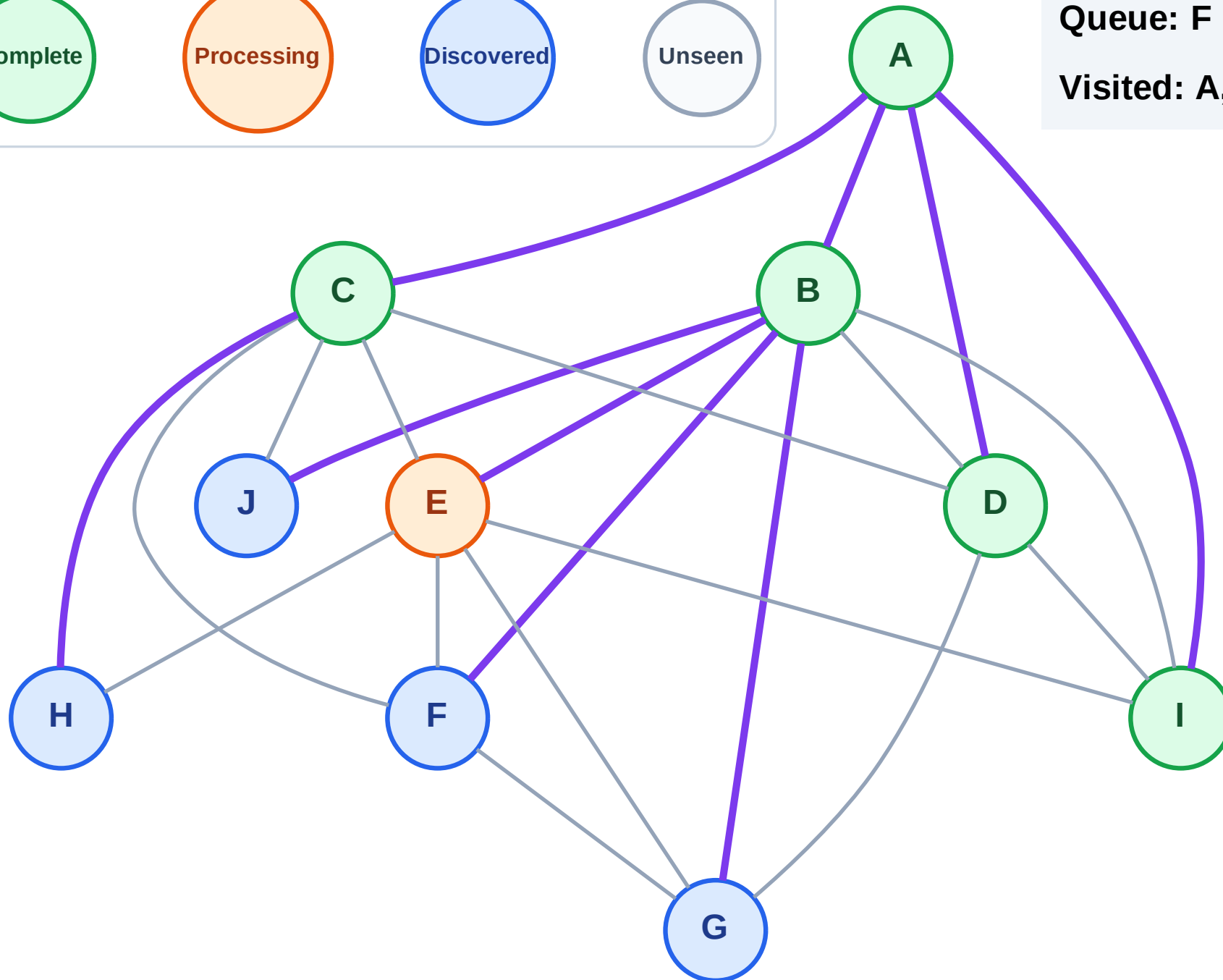
Step 12: Process node E

Legend



Queue: F → G → J → H

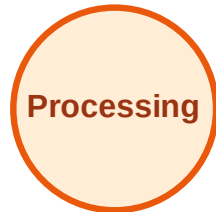
Visited: A, B, C, D, I



BFS Traversal

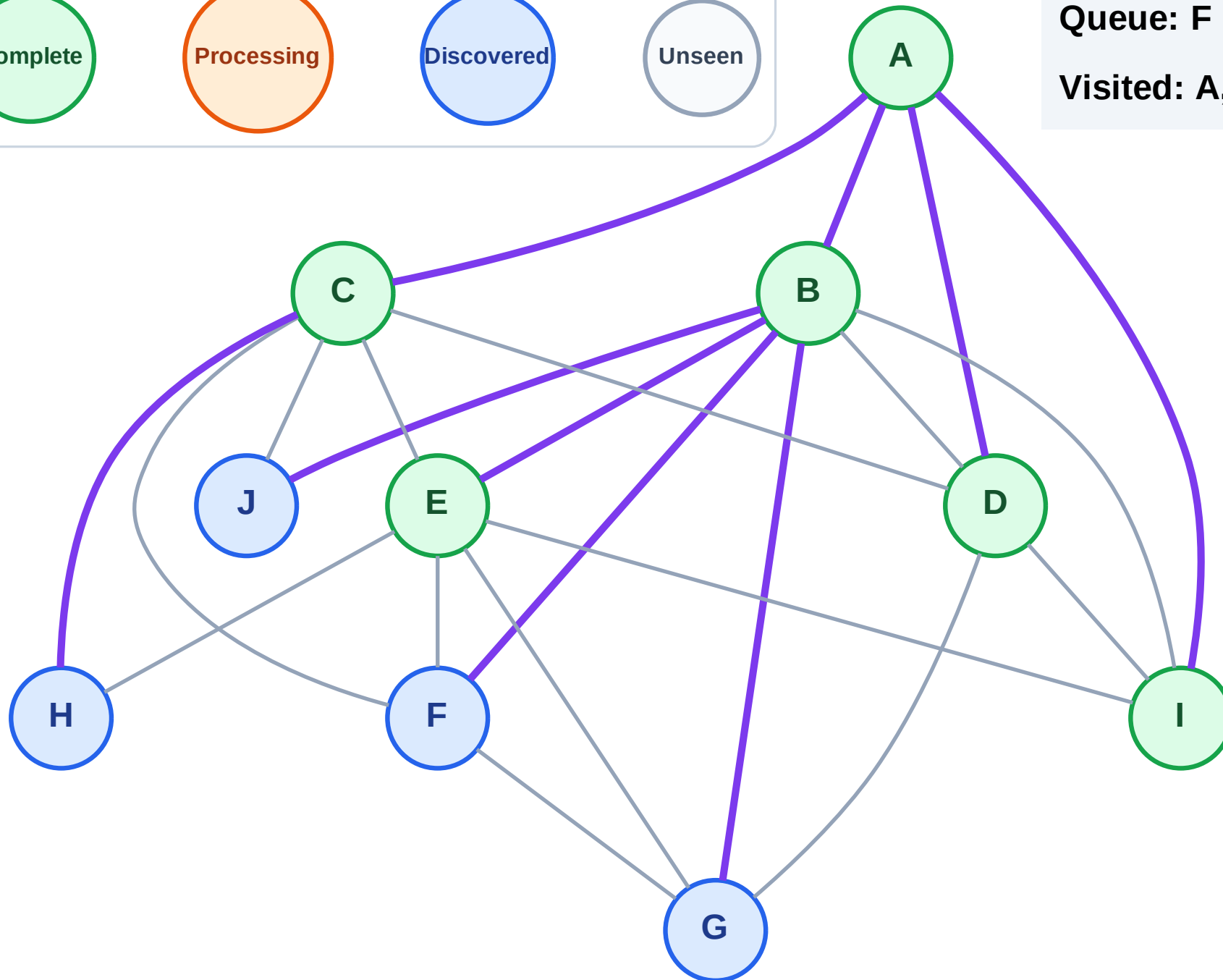
Step 13: No new neighbors from E

Legend



Queue: F → G → J → H

Visited: A, B, C, D, E, I



BFS Traversal

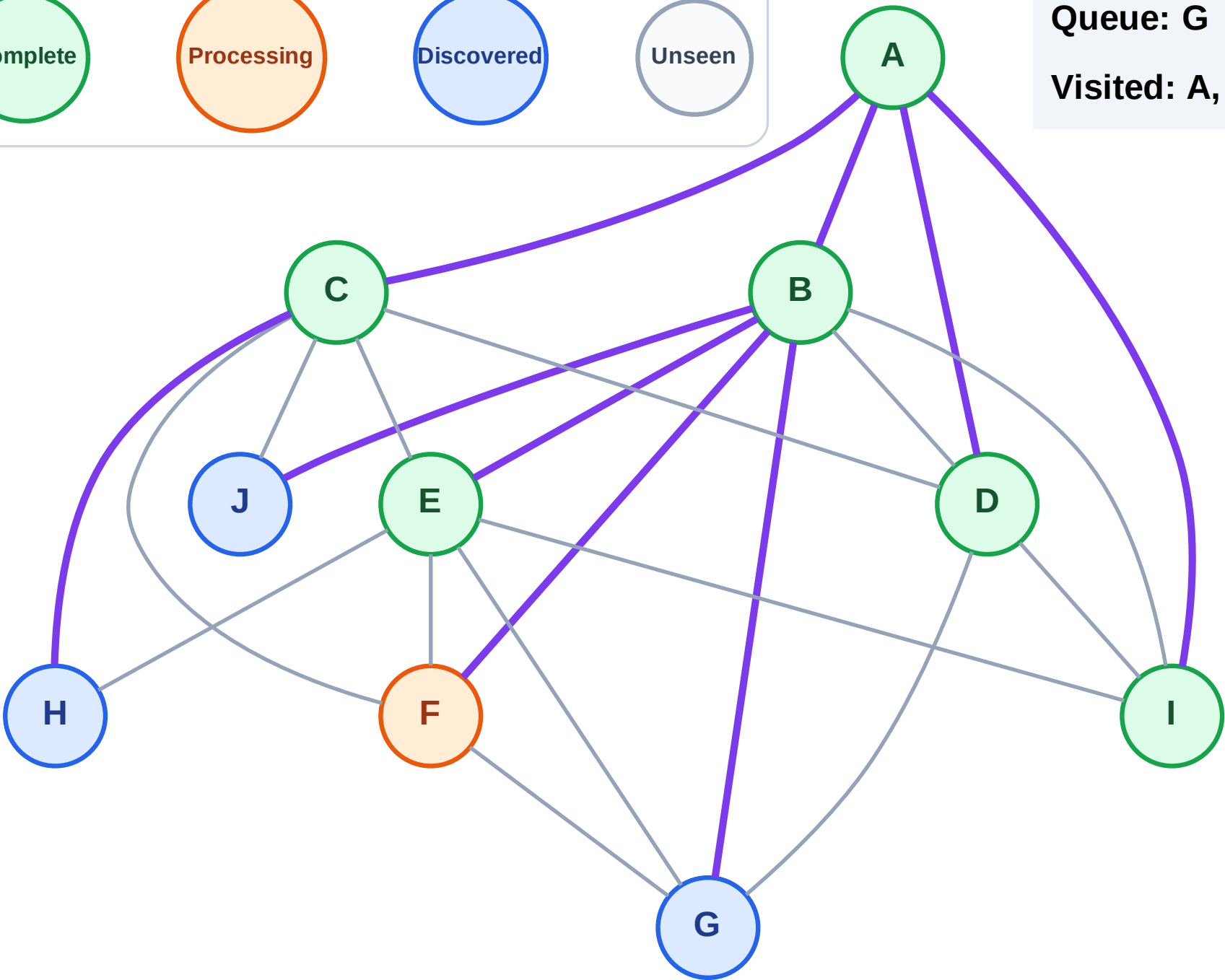
Step 14: Process node F

Legend



Queue: G → J → H

Visited: A, B, C, D, E, I



BFS Traversal

Step 15: No new neighbors from F

Legend

Complete

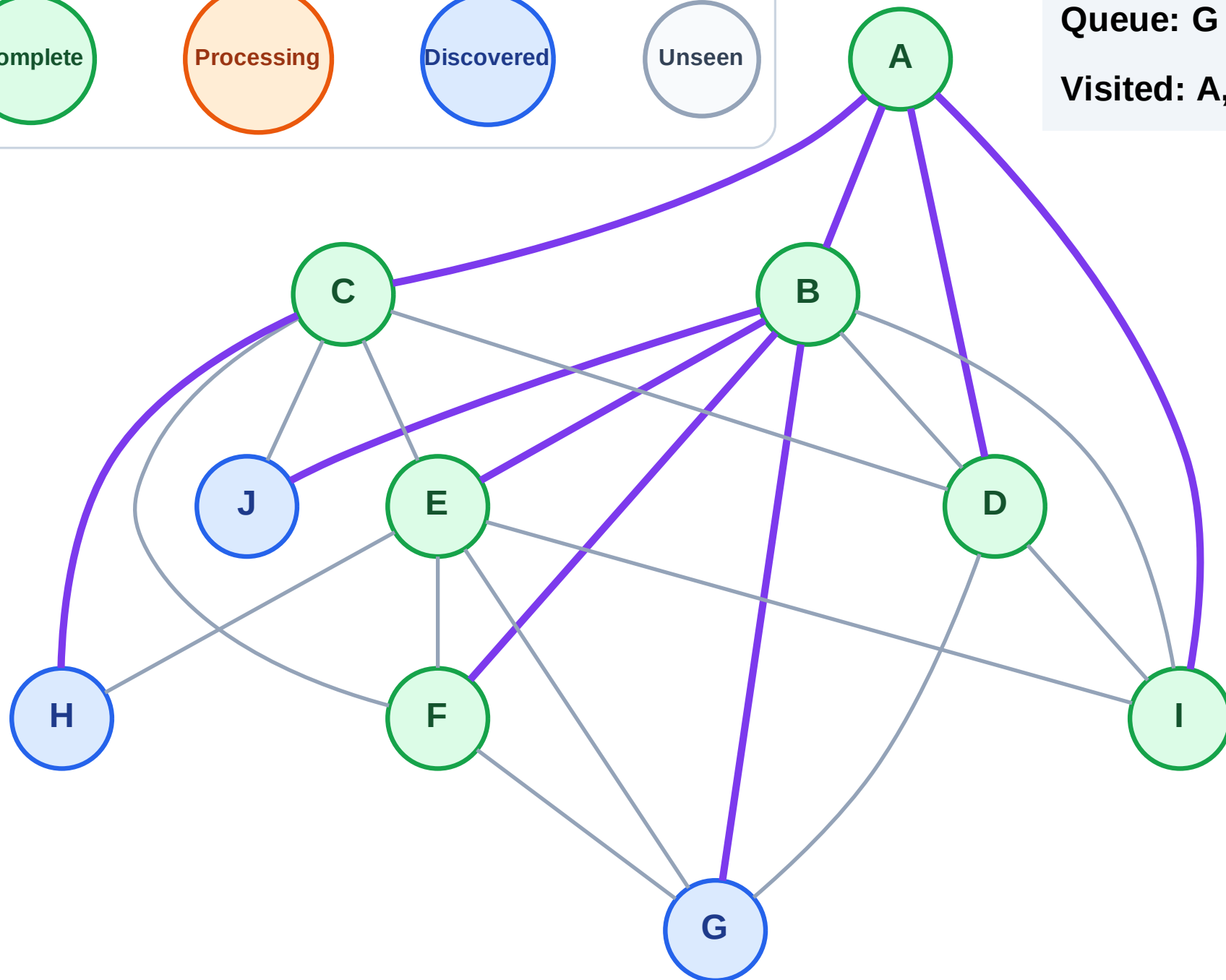
Processing

Discovered

Unseen

Queue: G → J → H

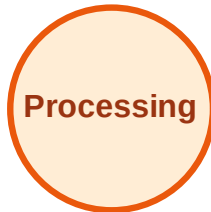
Visited: A, B, C, D, E, F, I



BFS Traversal

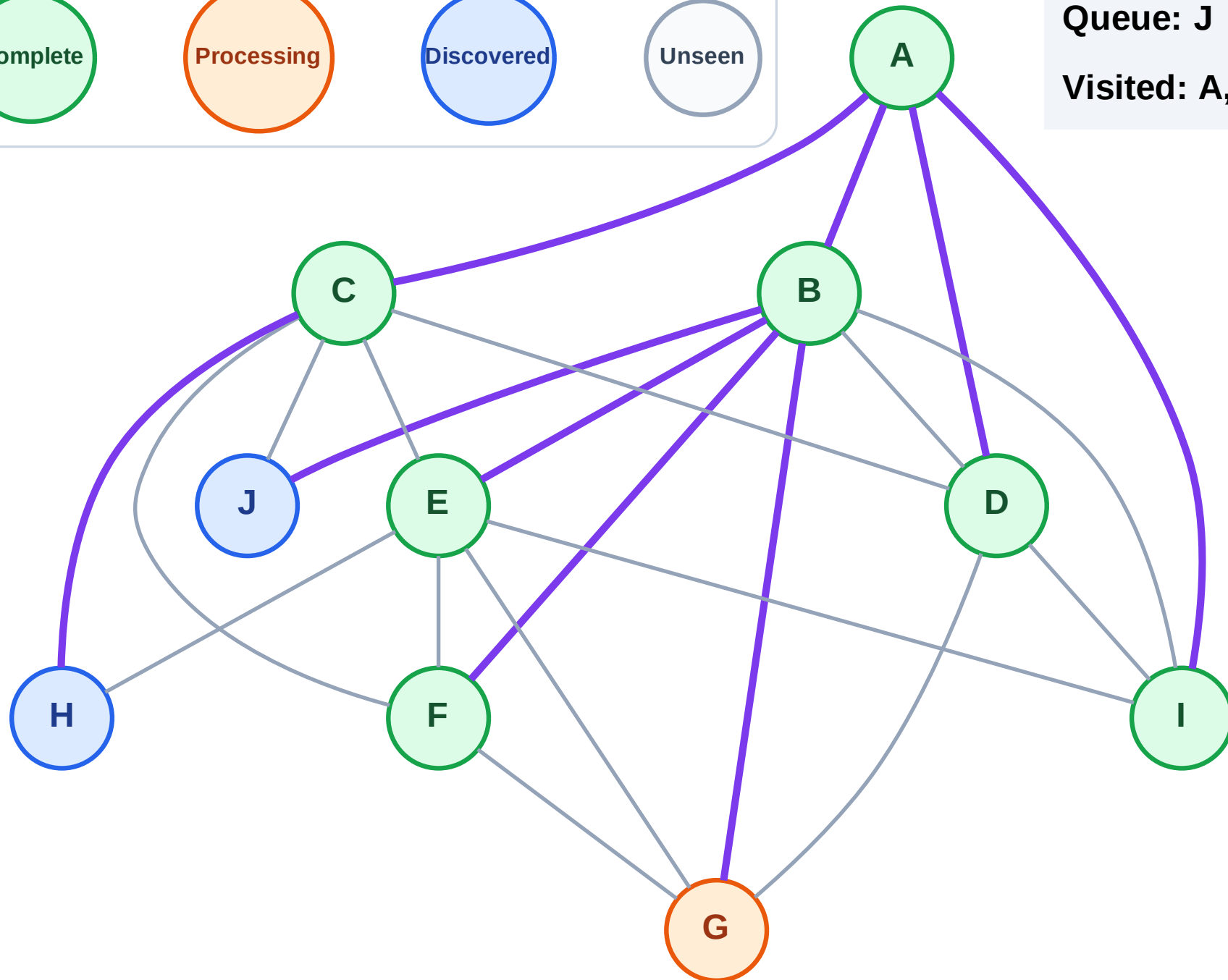
Step 16: Process node G

Legend



Queue: J → H

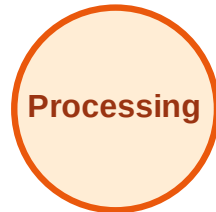
Visited: A, B, C, D, E, F, I



BFS Traversal

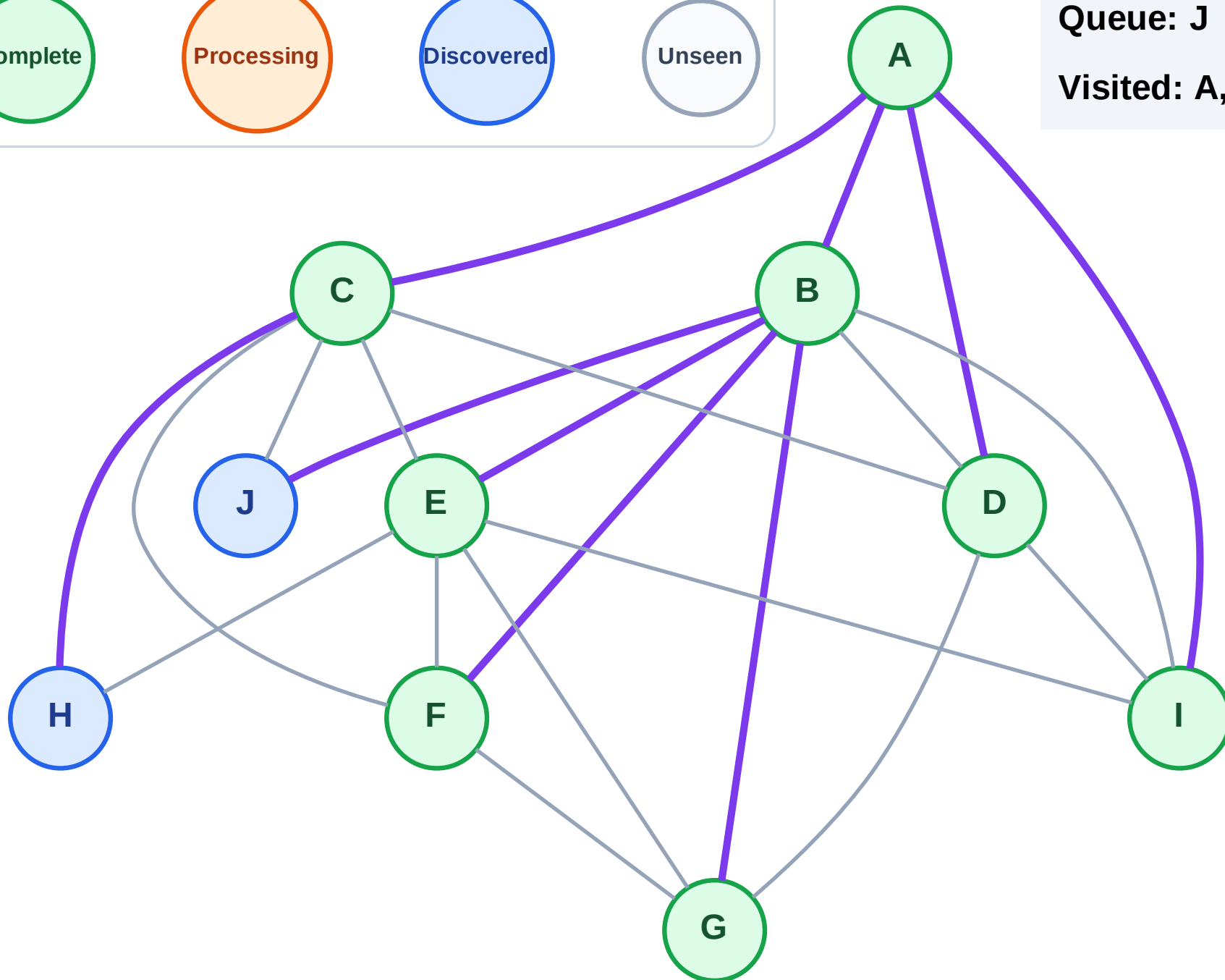
Step 17: No new neighbors from G

Legend



Queue: J → H

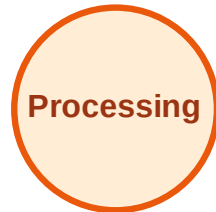
Visited: A, B, C, D, E, F, G, I



BFS Traversal

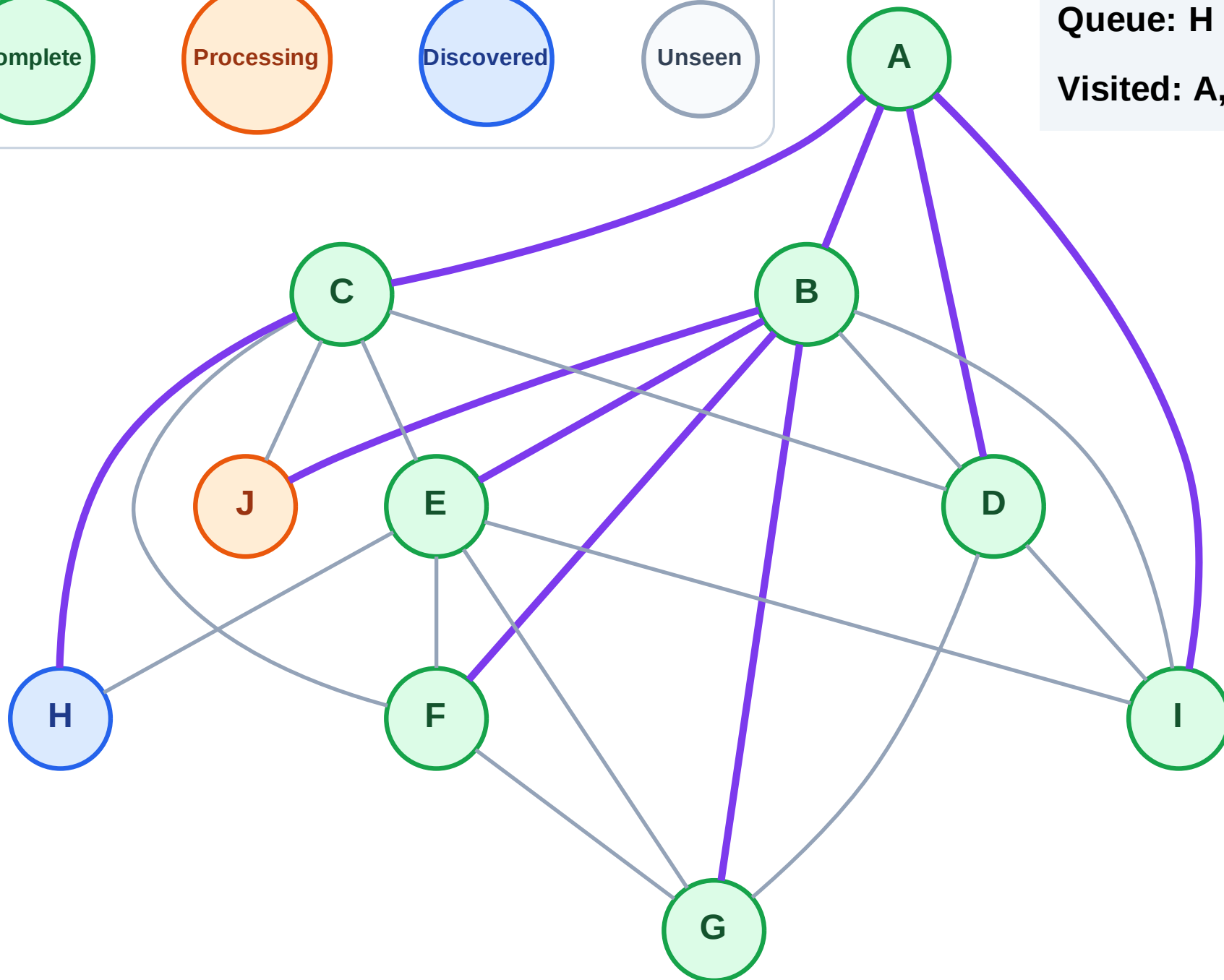
Step 18: Process node J

Legend



Queue: H

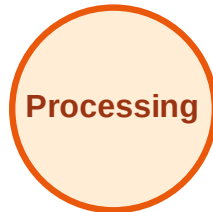
Visited: A, B, C, D, E, F, G, I



BFS Traversal

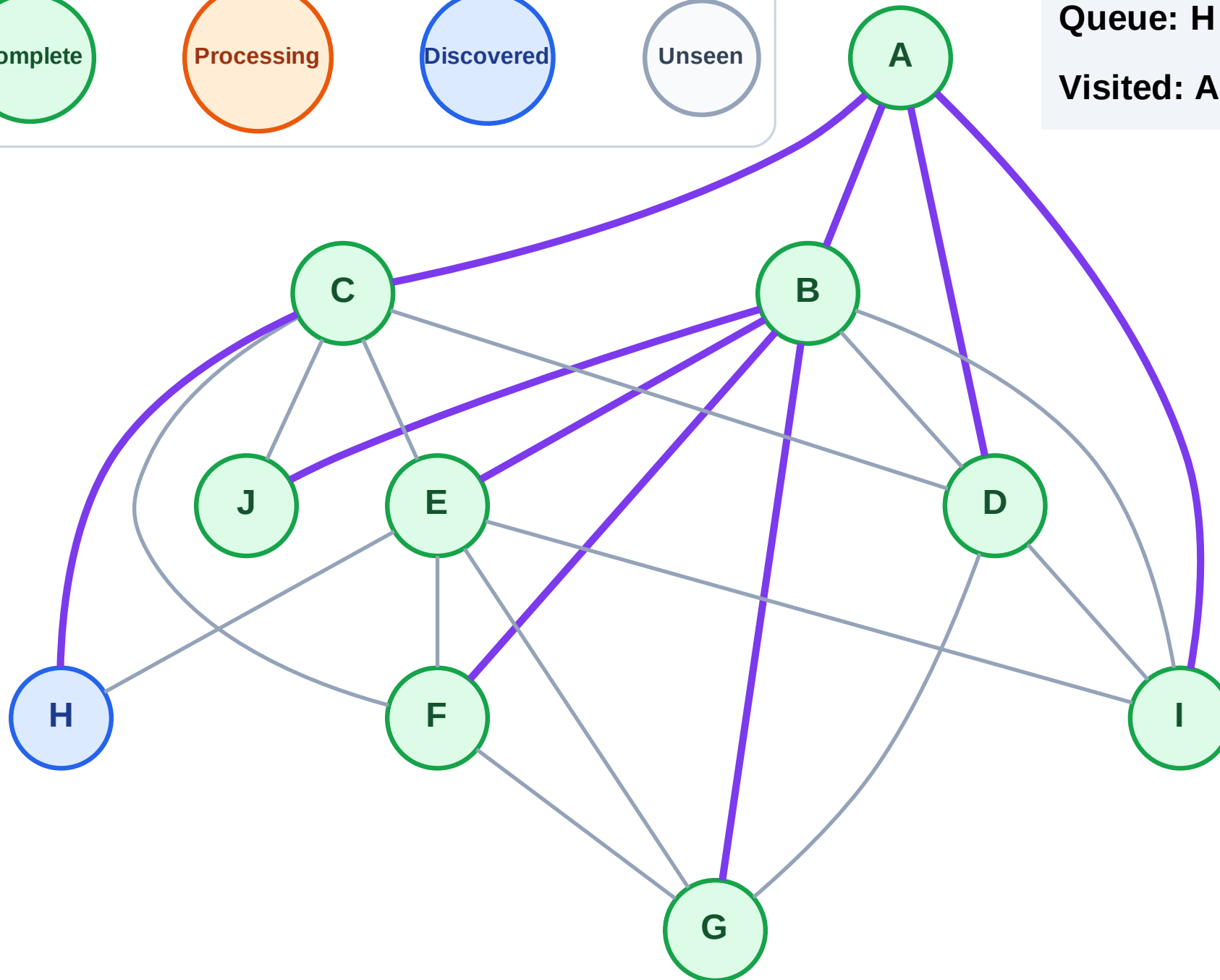
Step 19: No new neighbors from J

Legend



Queue: H

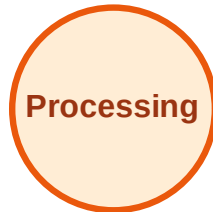
Visited: A, B, C, D, E, F, G, I, J



BFS Traversal

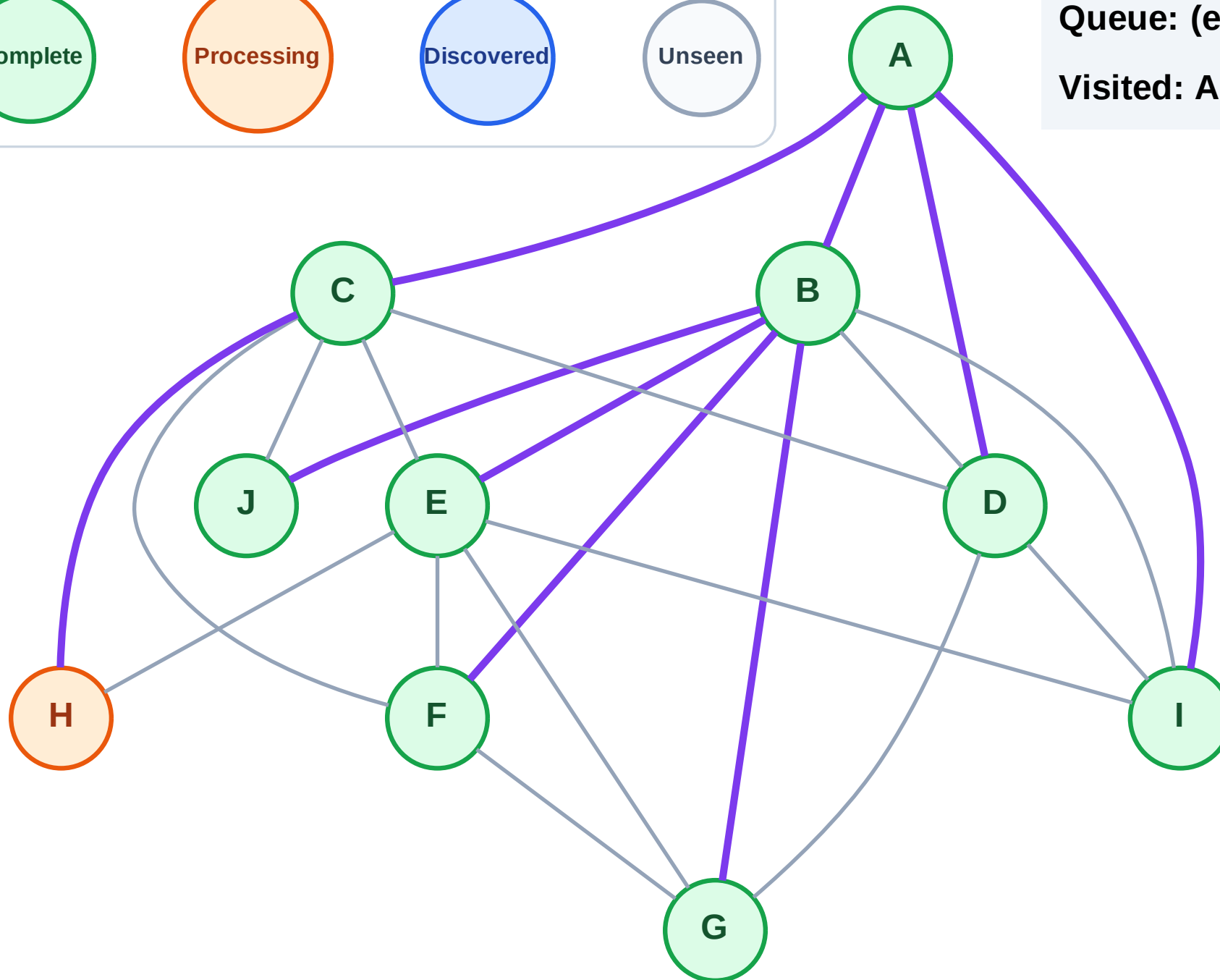
Step 20: Process node H

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, I, J



BFS Traversal

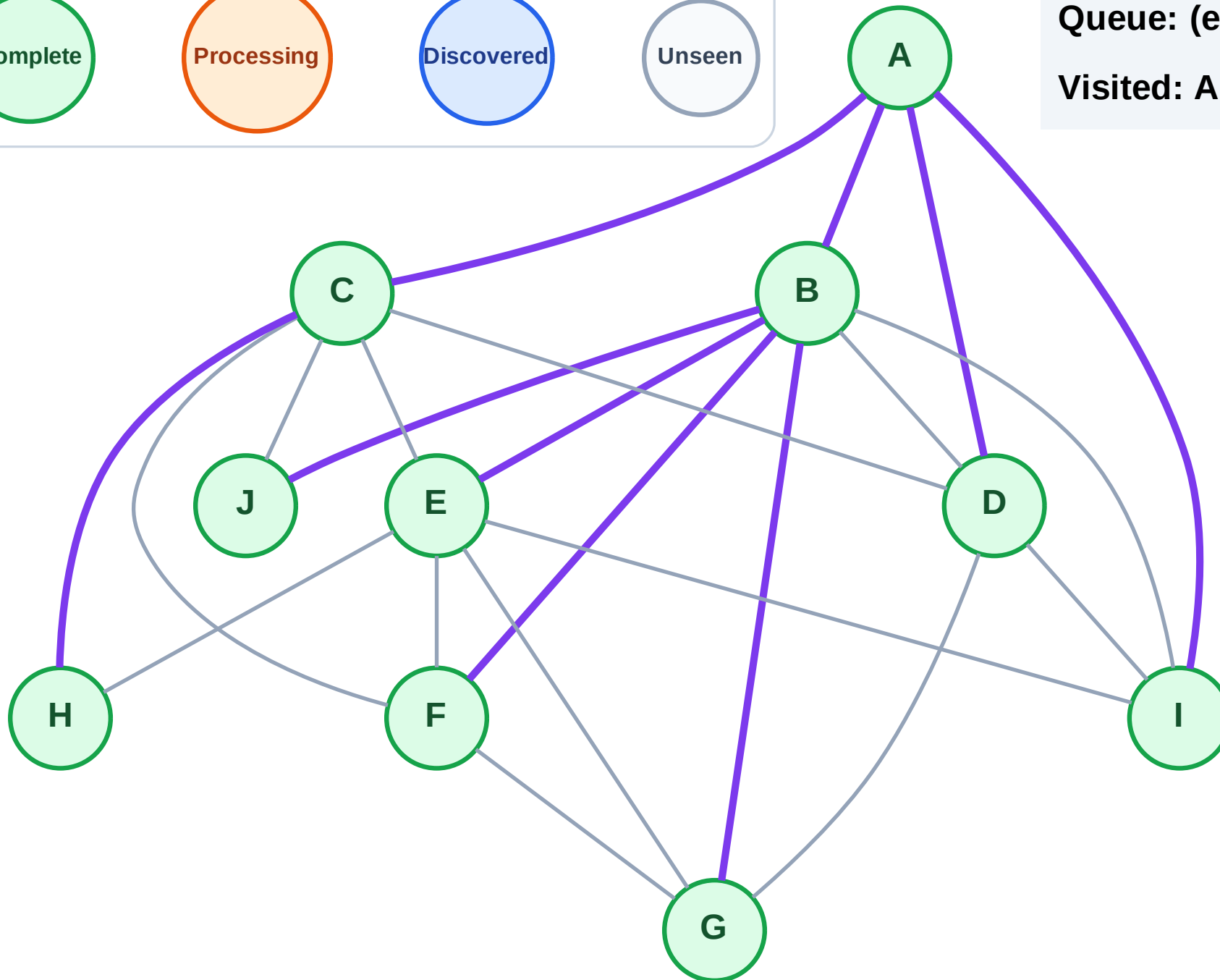
Step 21: No new neighbors from H

Legend



Queue: (empty)

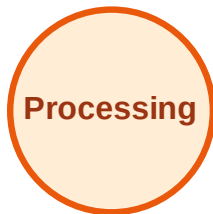
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

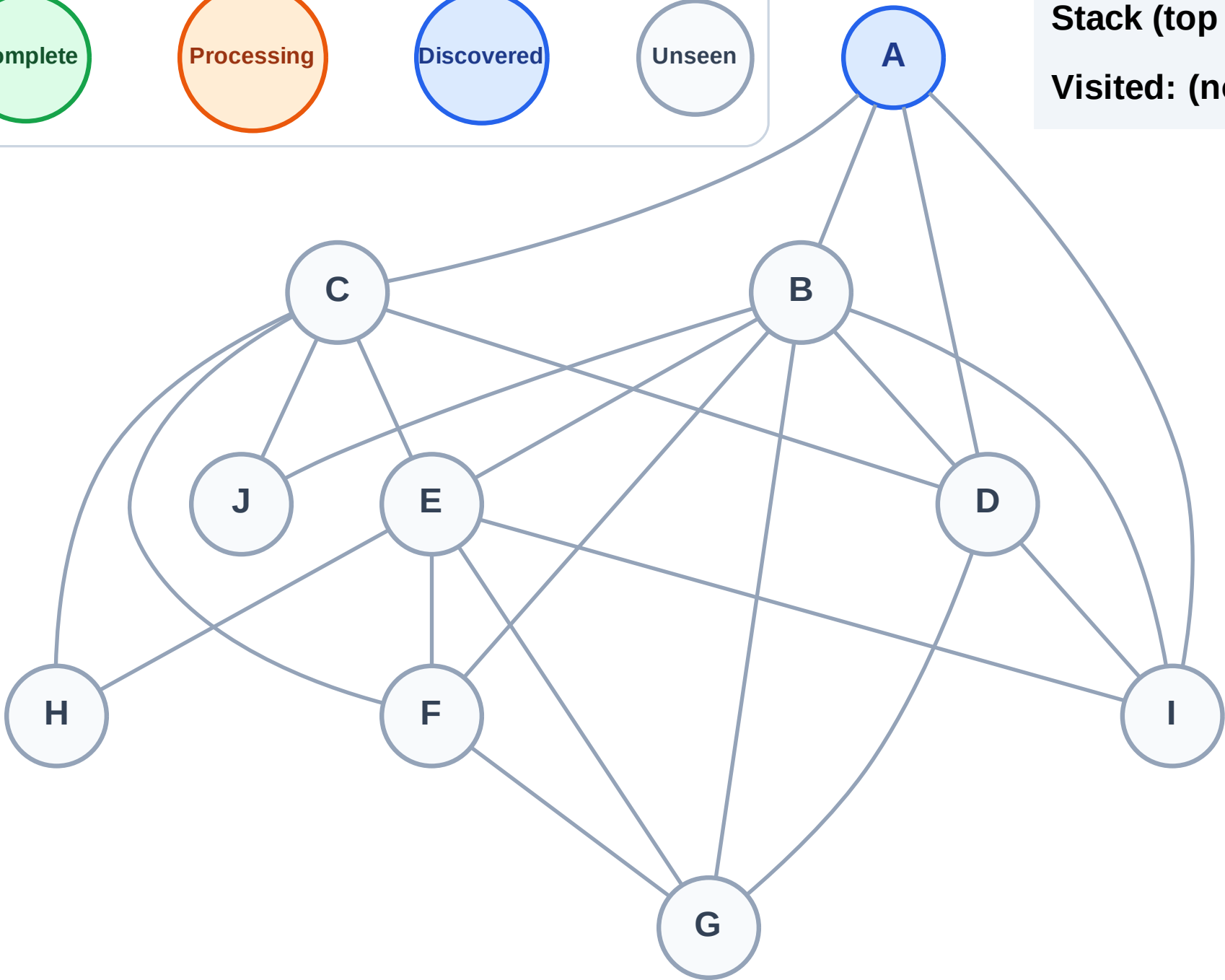
Step 1: Start at A

Legend



Stack (top at right): A

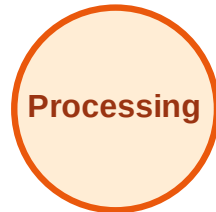
Visited: (none)



DFS Traversal

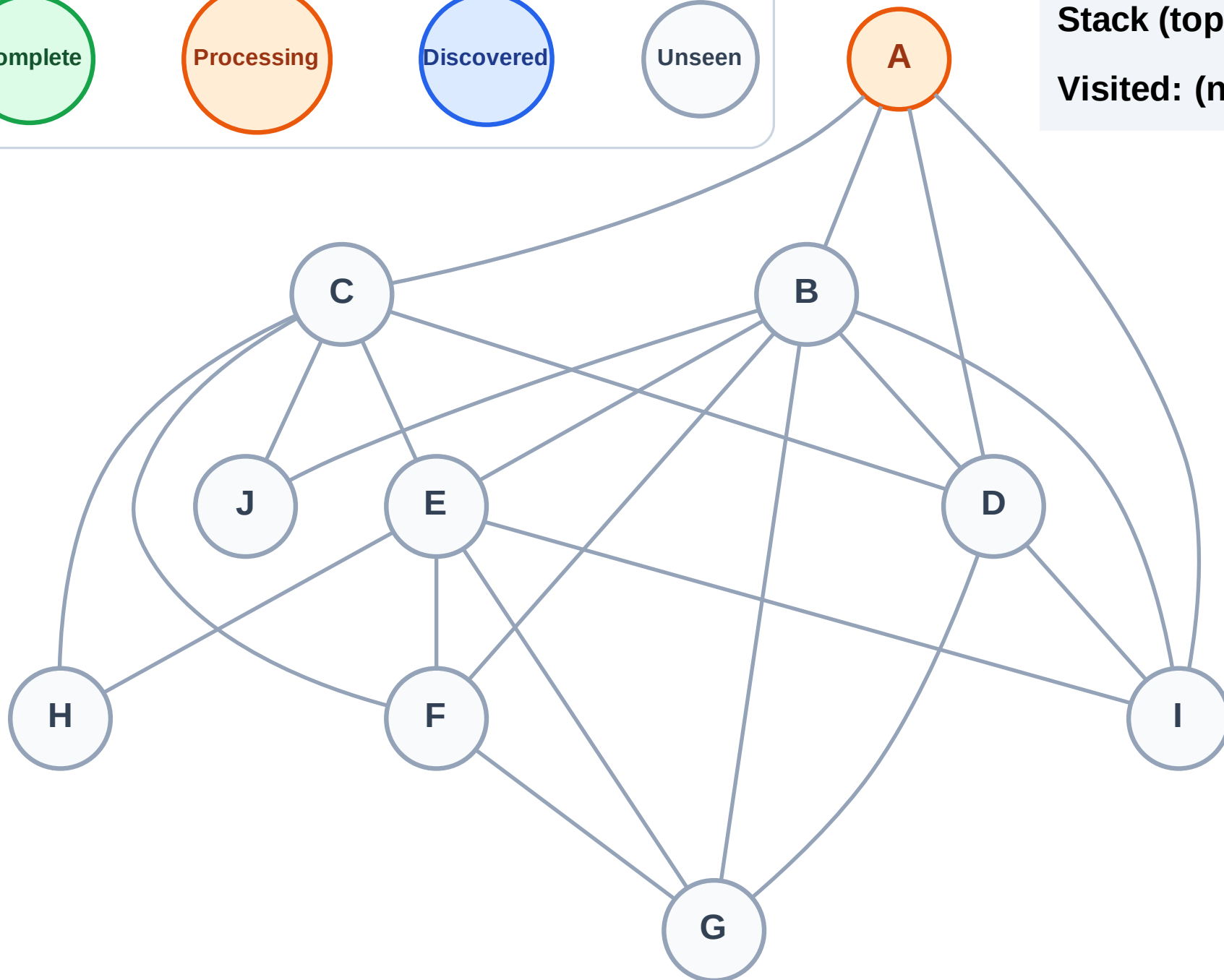
Step 2: Process node A

Legend



Stack (top at right): (empty)

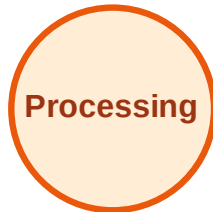
Visited: (none)



DFS Traversal

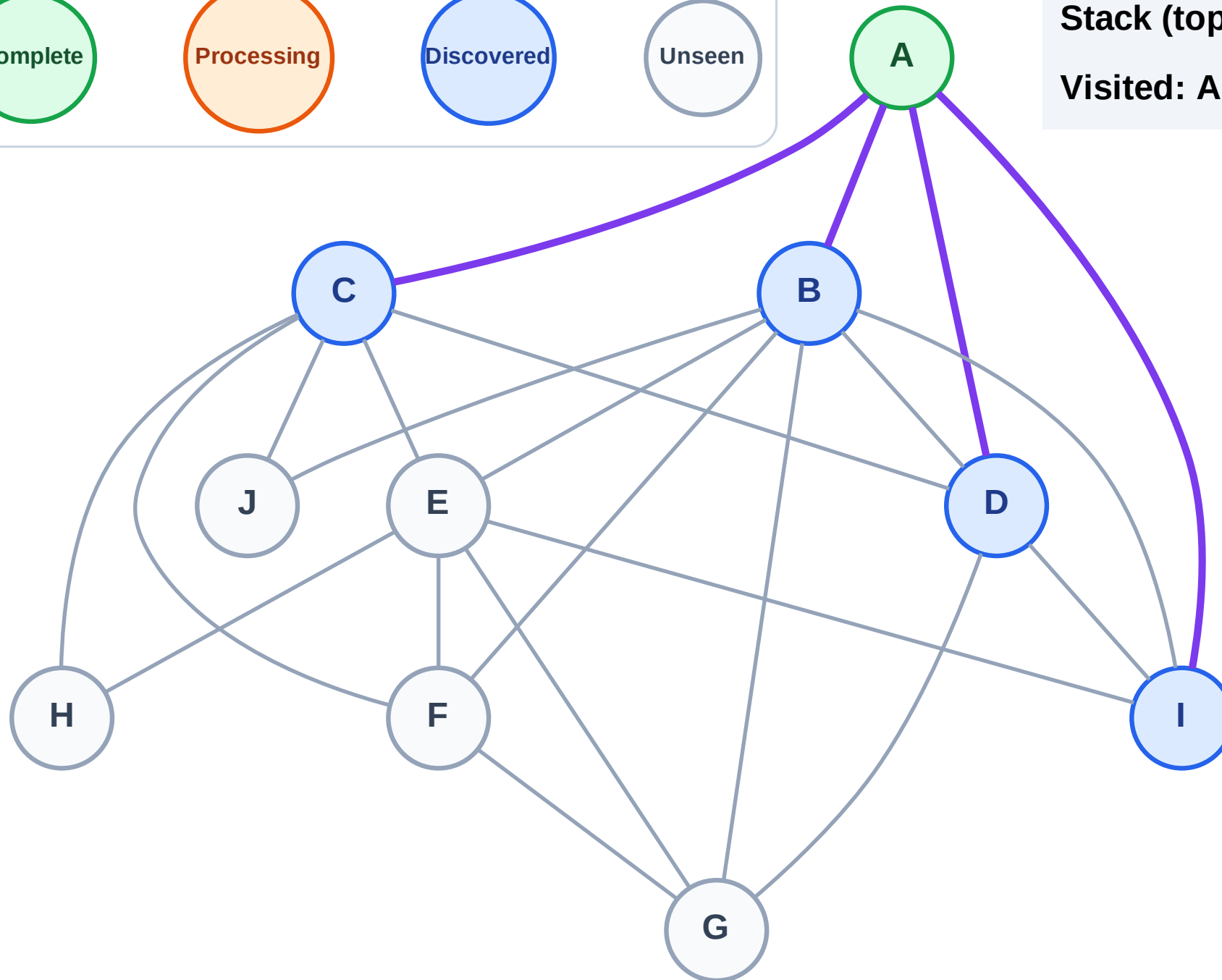
Step 3: Discover I, D, C, B from A

Legend



Stack (top at right): I | D | C | B

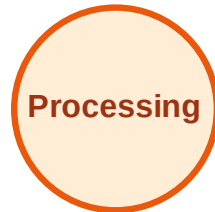
Visited: A



DFS Traversal

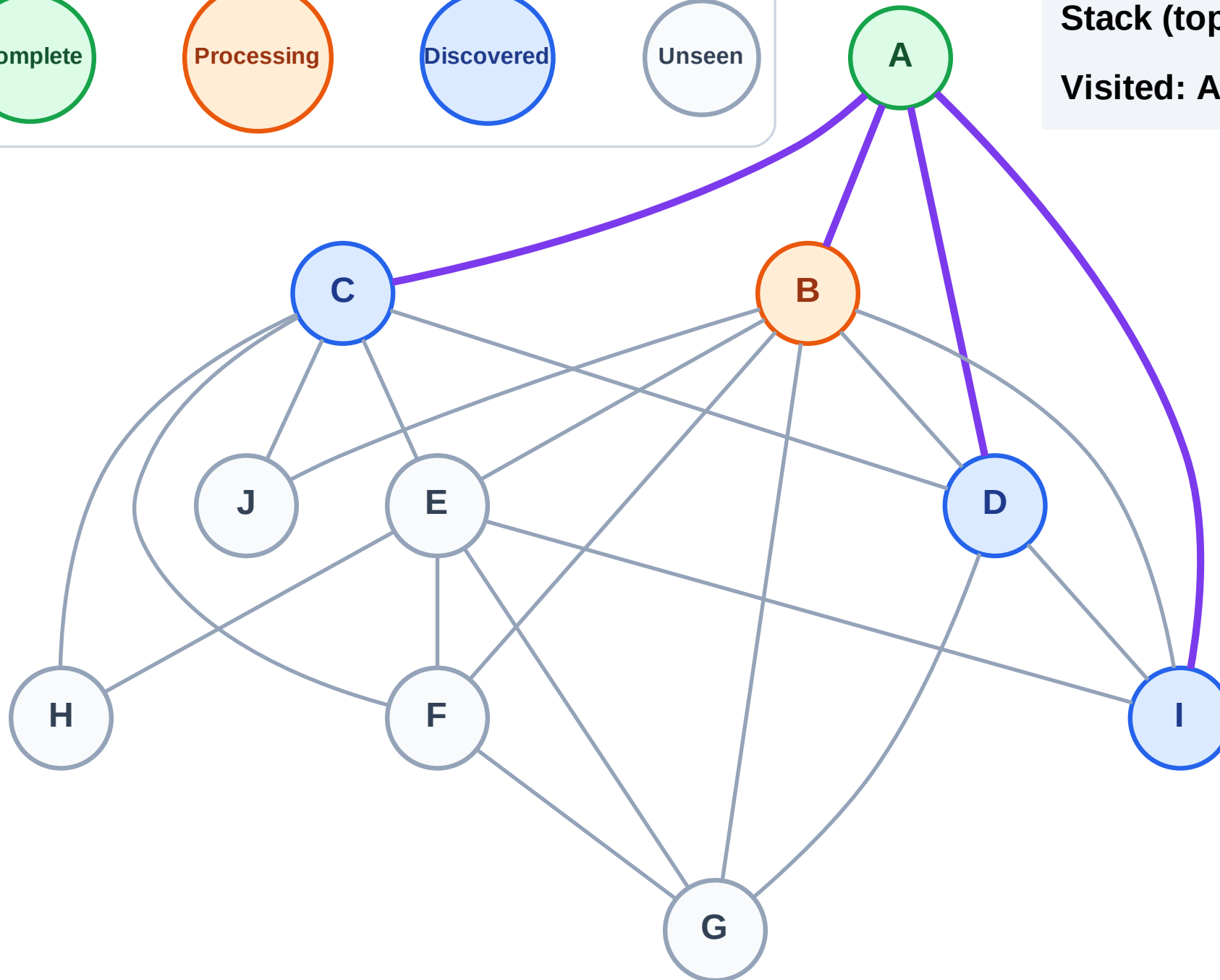
Step 4: Process node B

Legend



Stack (top at right): I | D | C

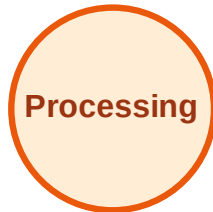
Visited: A



DFS Traversal

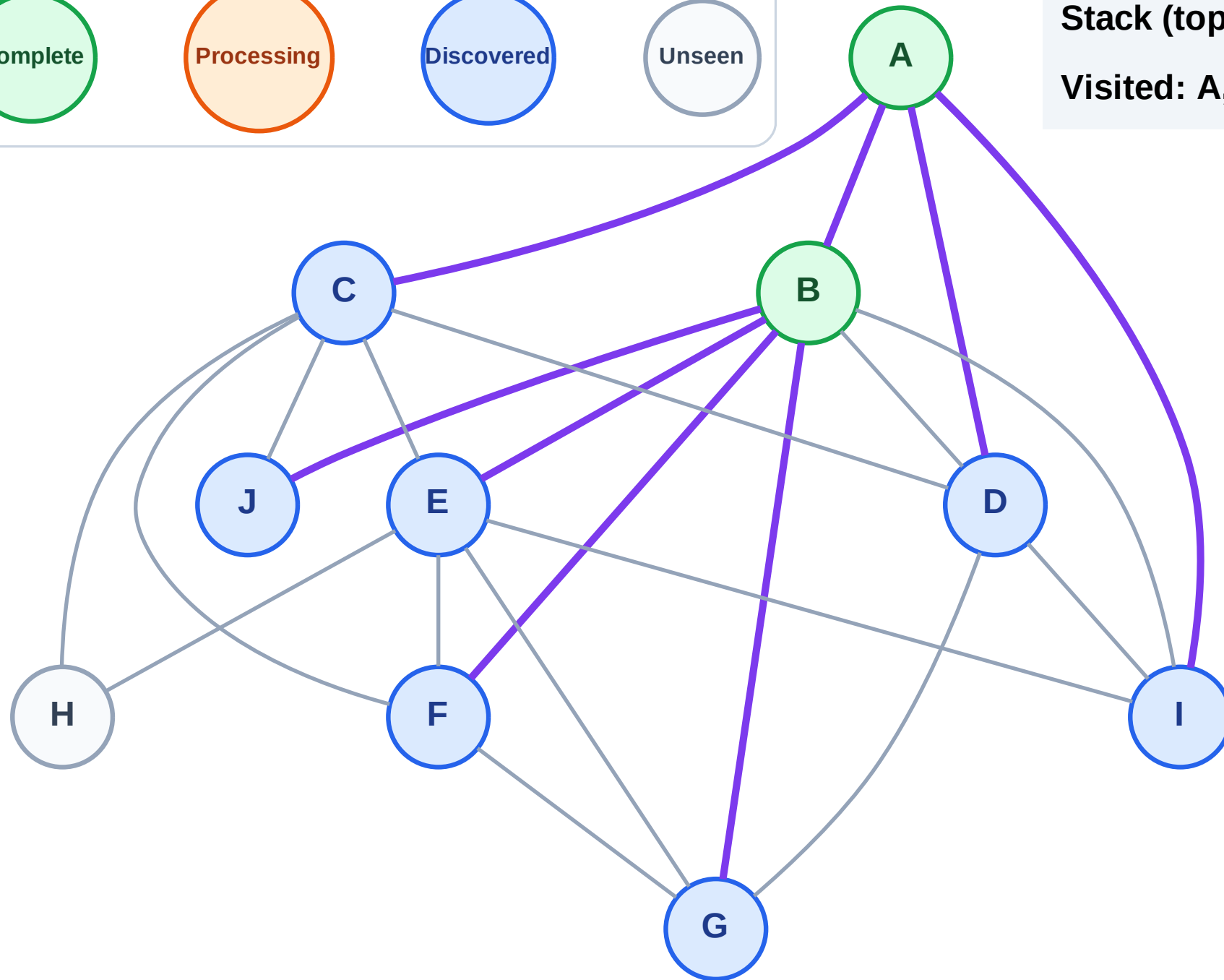
Step 5: Discover J, G, F, E from B

Legend



Stack (top at right): I | D | C | J | G | F | E

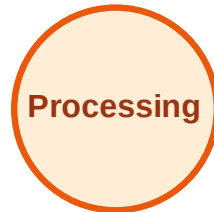
Visited: A, B



DFS Traversal

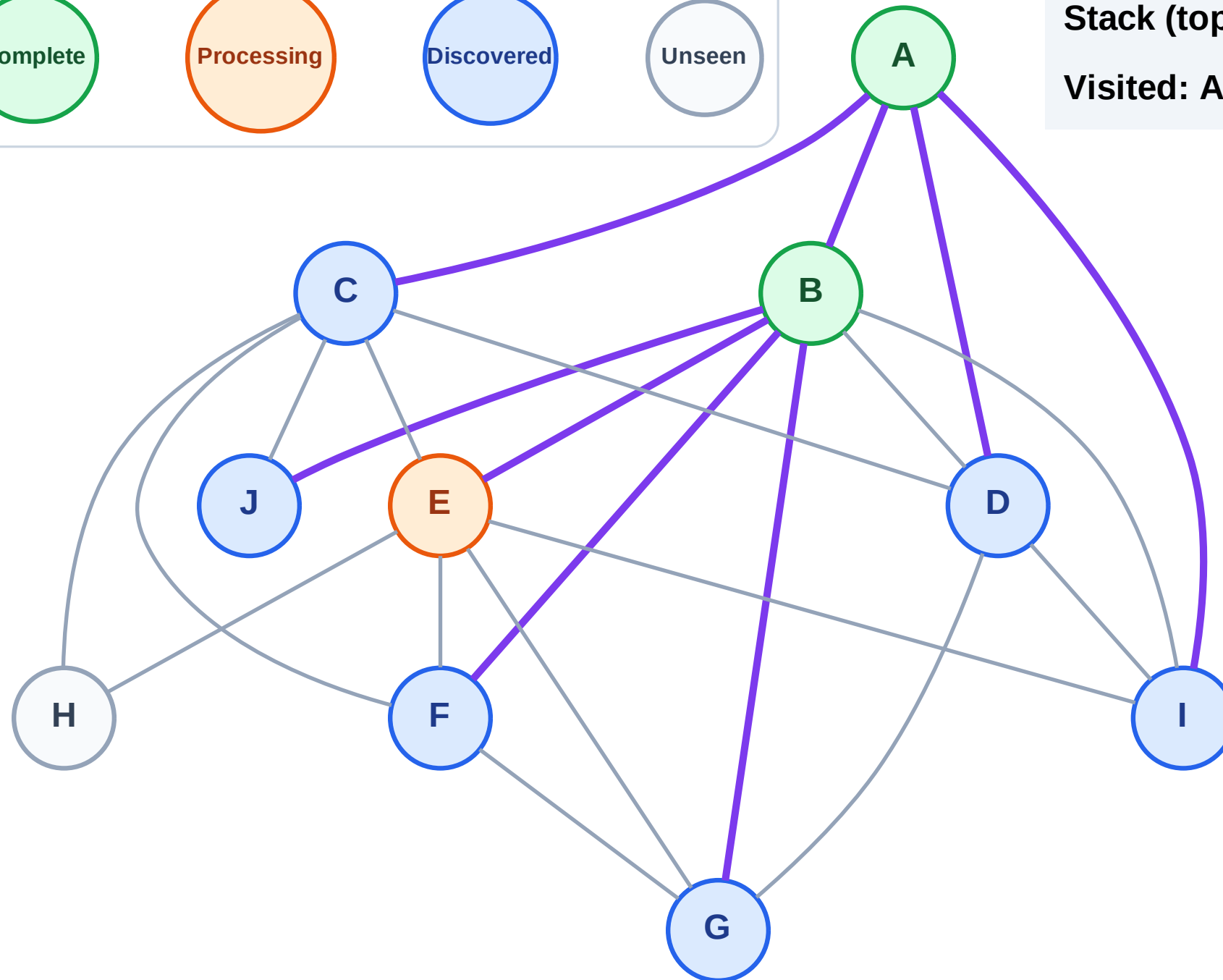
Step 6: Process node E

Legend



Stack (top at right): I | D | C | J | G | F

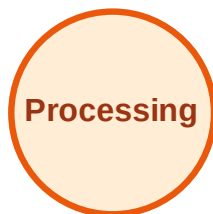
Visited: A, B



DFS Traversal

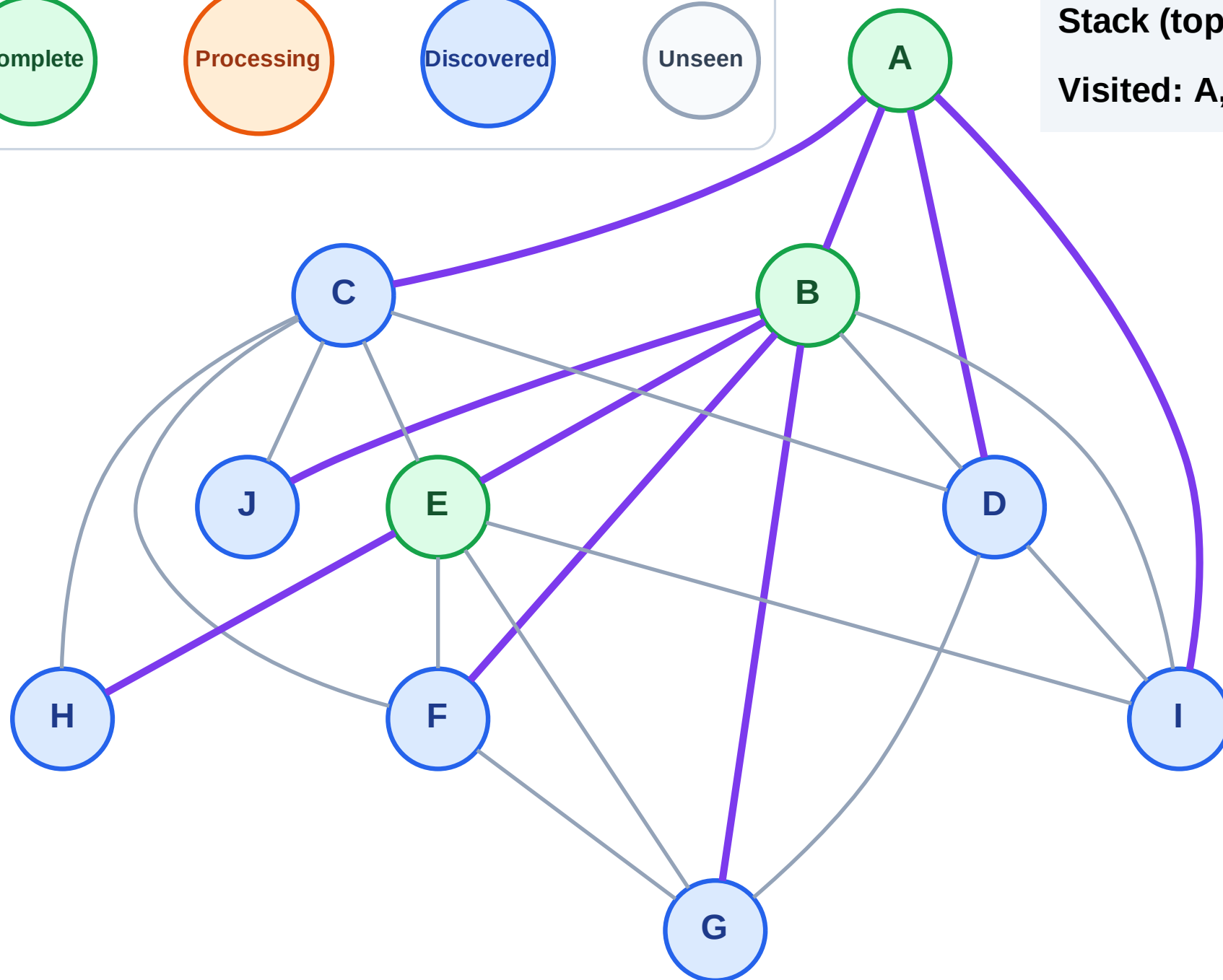
Step 7: Discover H from E

Legend



Stack (top at right): I | D | C | J | G | F | H

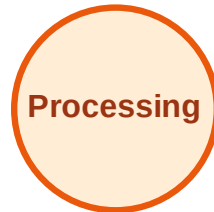
Visited: A, B, E



DFS Traversal

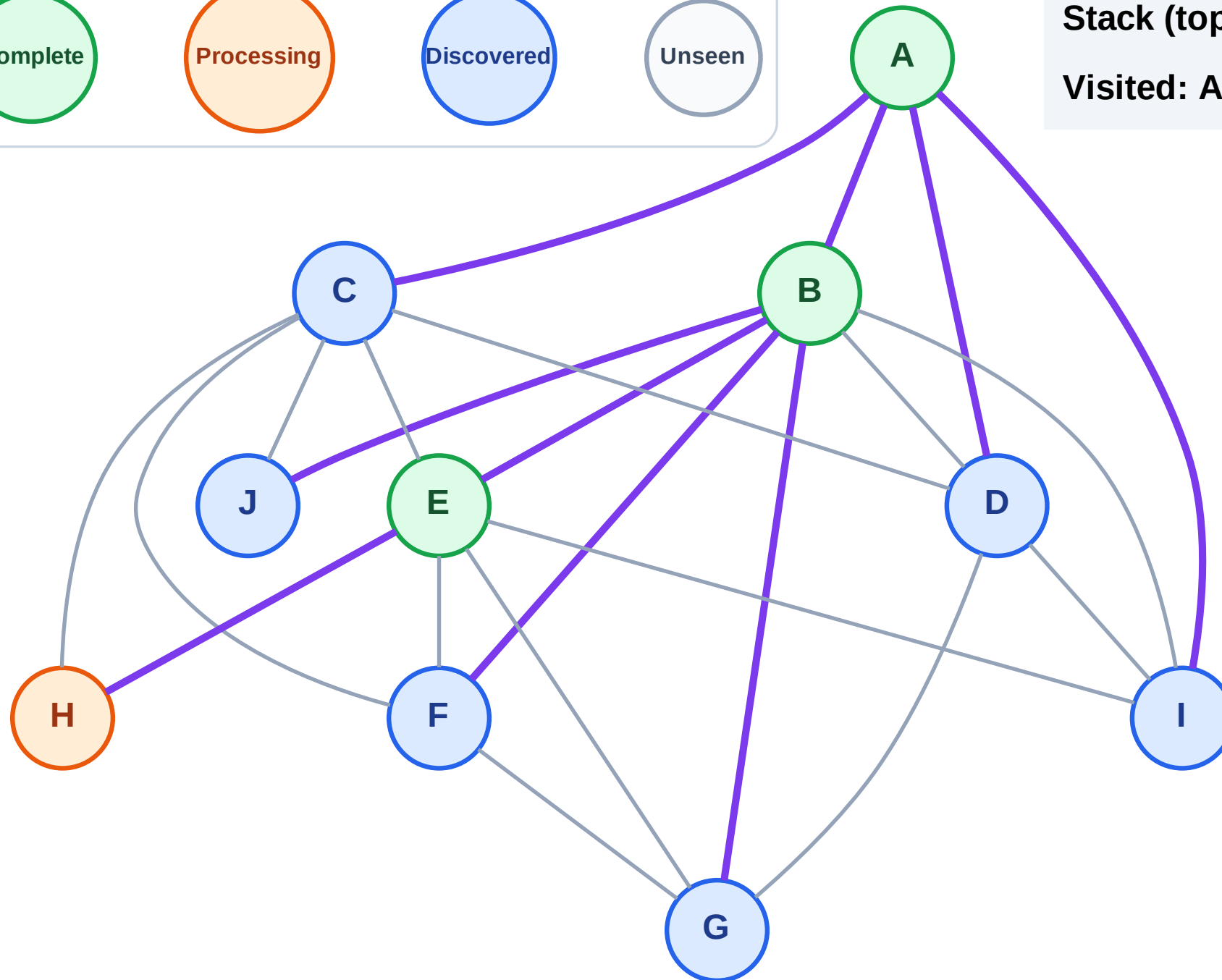
Step 8: Process node H

Legend



Stack (top at right): I | D | C | J | G | F

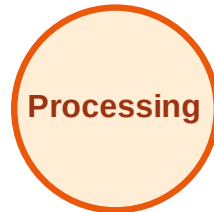
Visited: A, B, E



DFS Traversal

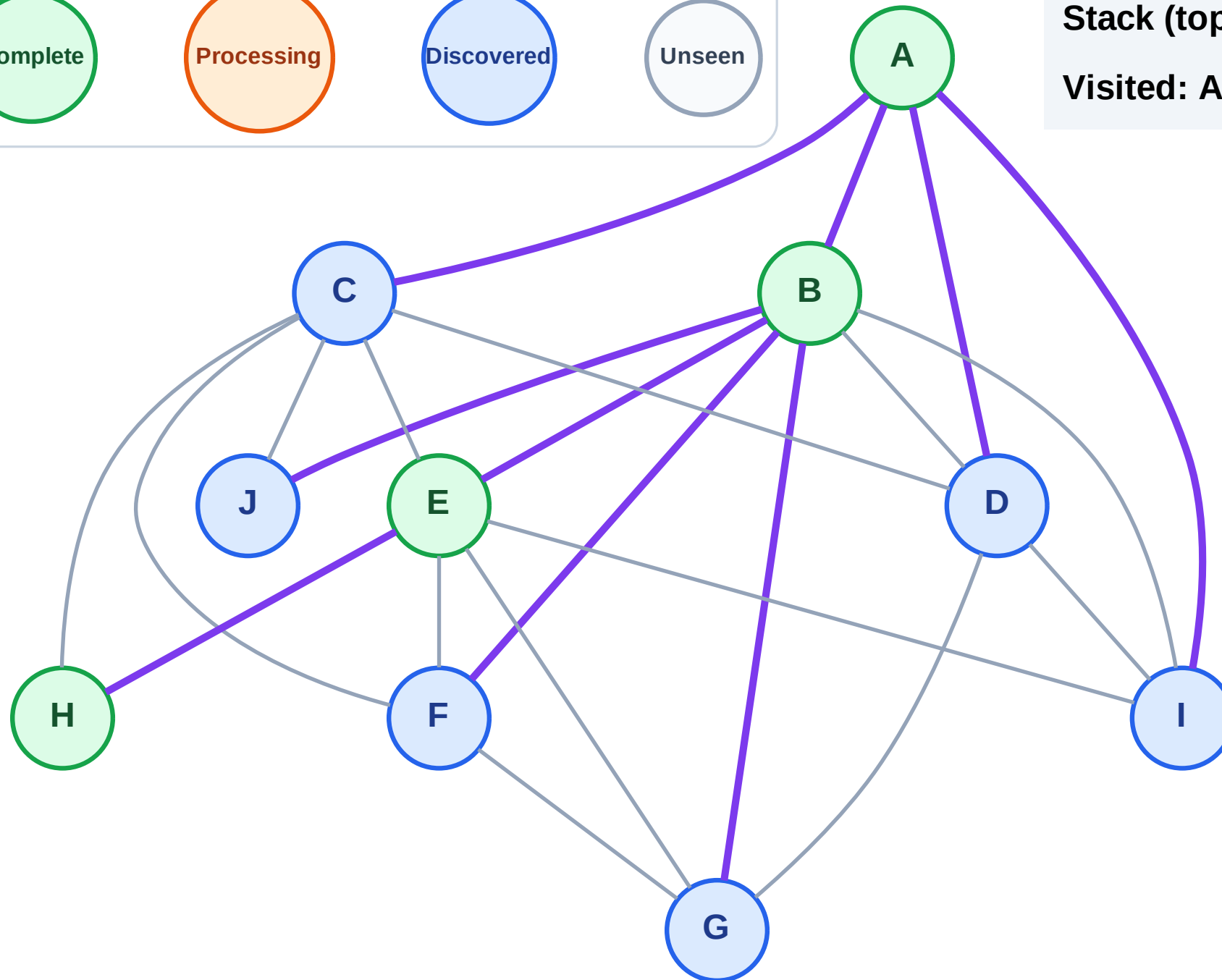
Step 9: No new neighbors from H

Legend



Stack (top at right): I | D | C | J | G | F

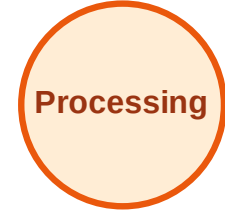
Visited: A, B, E, H



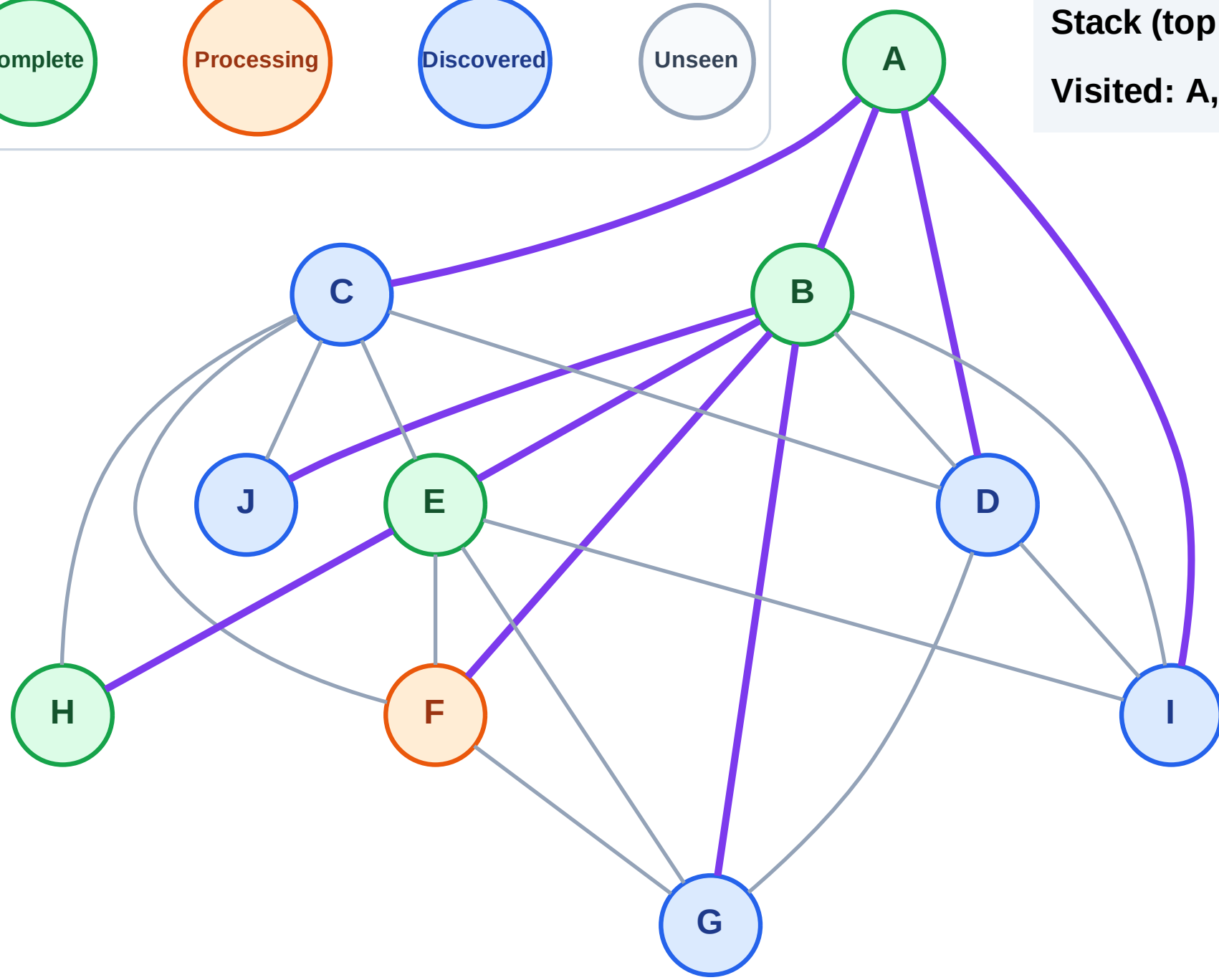
DFS Traversal

Step 10: Process node F

Legend



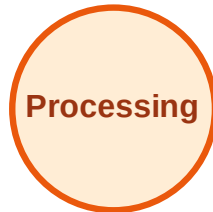
Stack (top at right): I | D | C | J | G
Visited: A, B, E, H



DFS Traversal

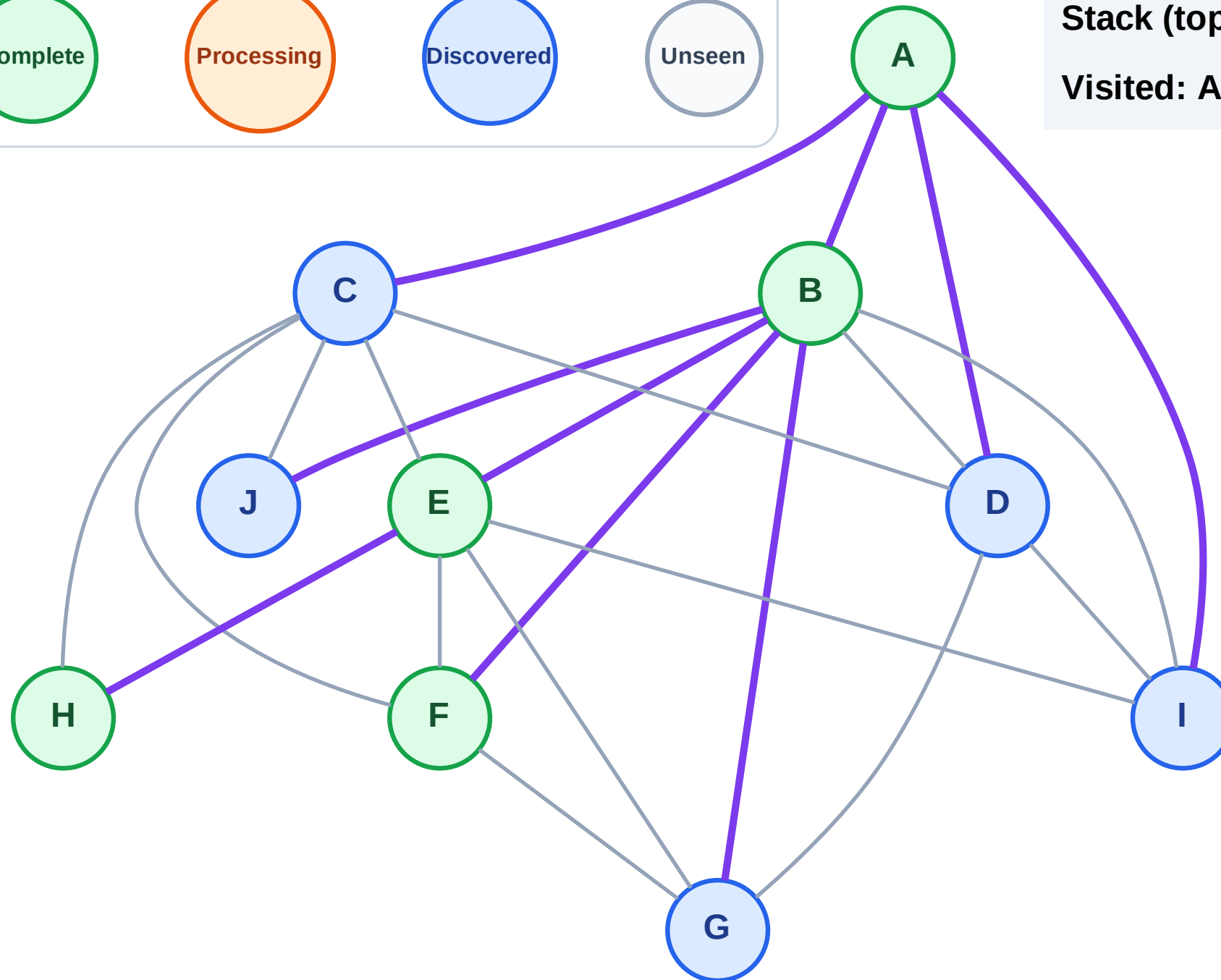
Step 11: No new neighbors from F

Legend



Stack (top at right): I | D | C | J | G

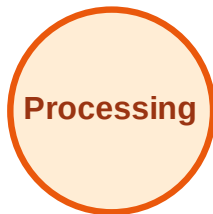
Visited: A, B, E, F, H



DFS Traversal

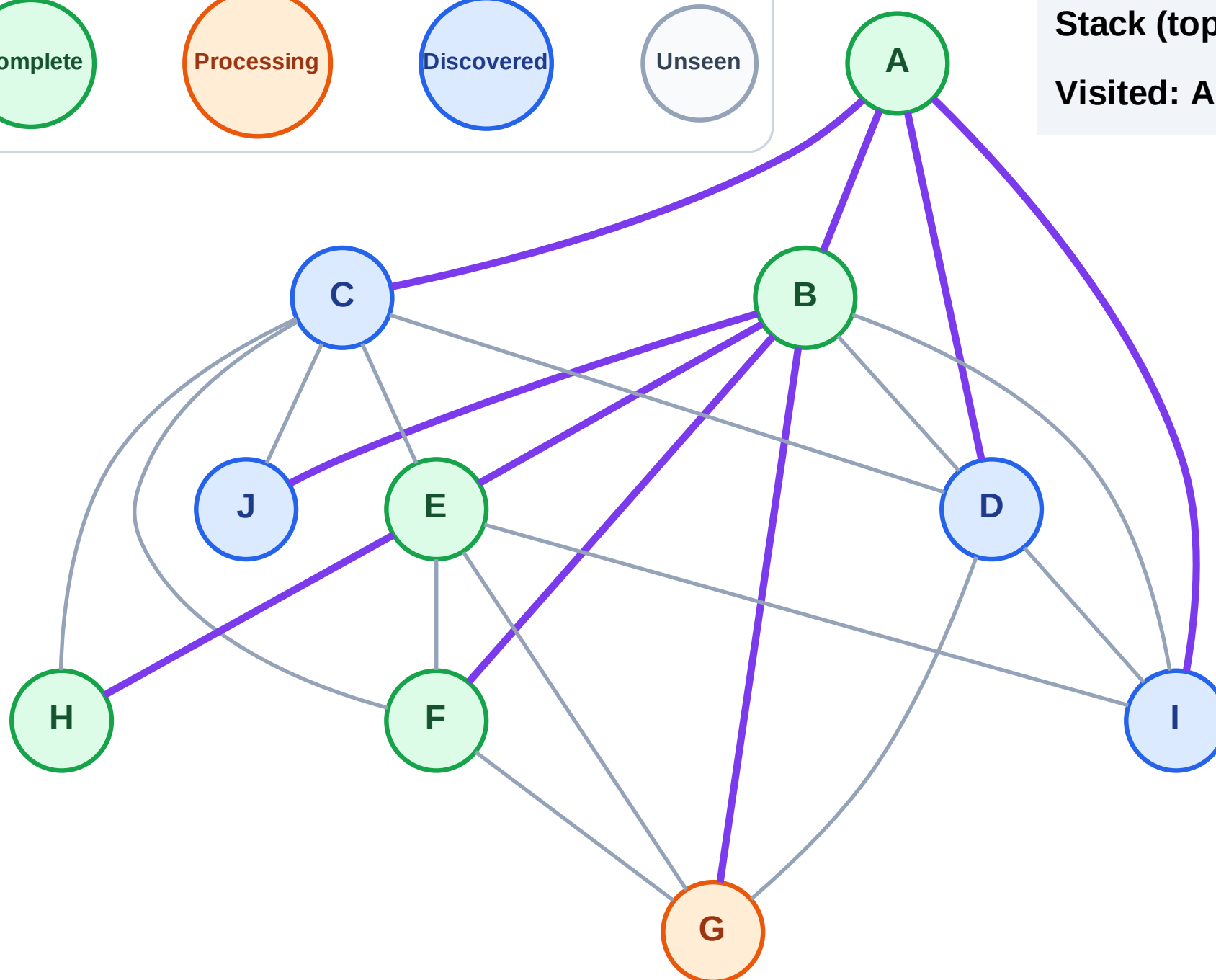
Step 12: Process node G

Legend



Stack (top at right): I | D | C | J

Visited: A, B, E, F, H



Step 13: No new neighbors from G

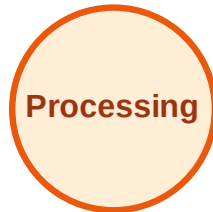
Step 13: No new neighbors from G



DFS Traversal

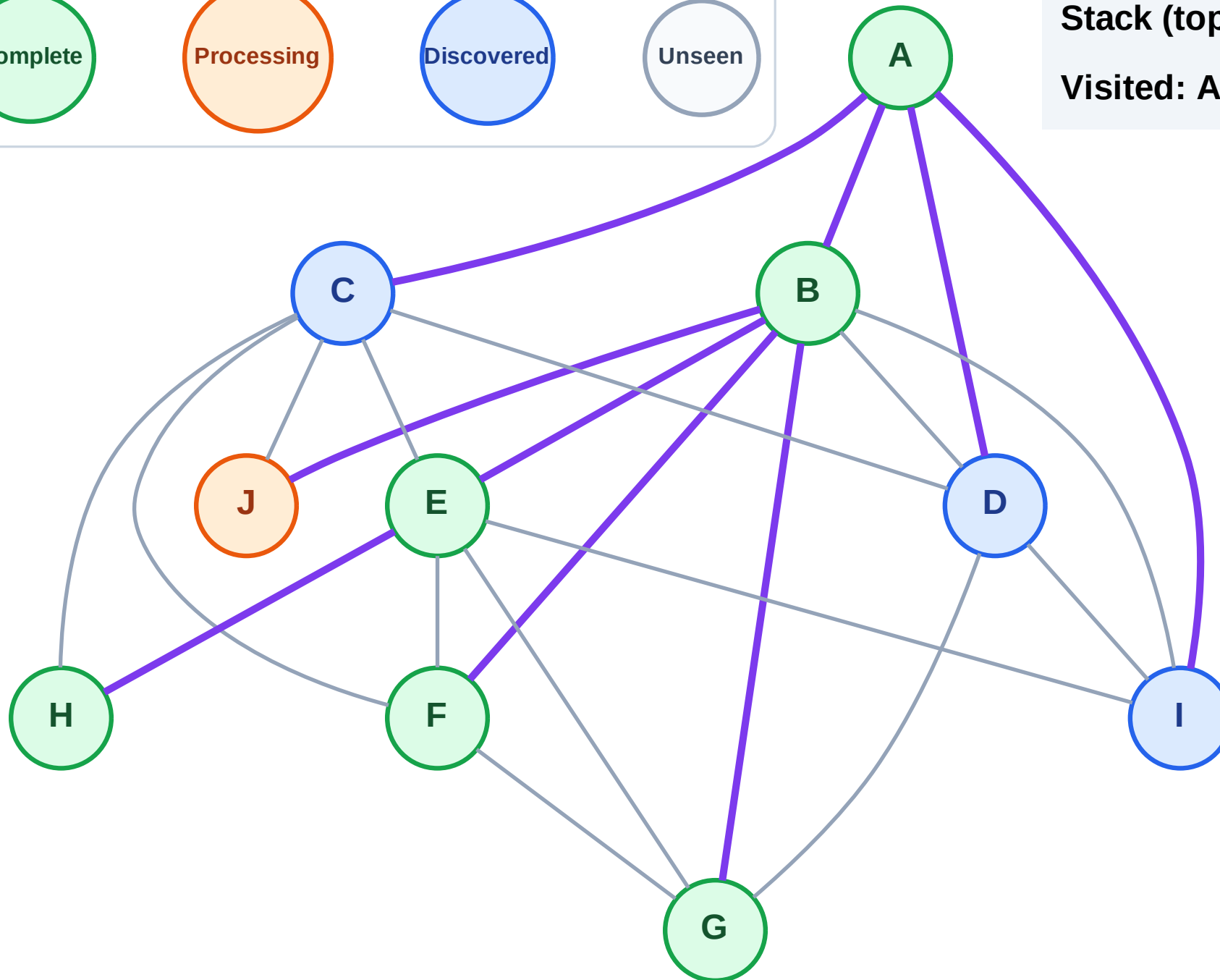
Step 14: Process node J

Legend



Stack (top at right): I | D | C

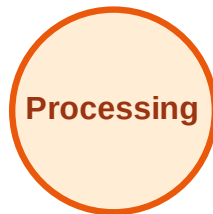
Visited: A, B, E, F, G, H



DFS Traversal

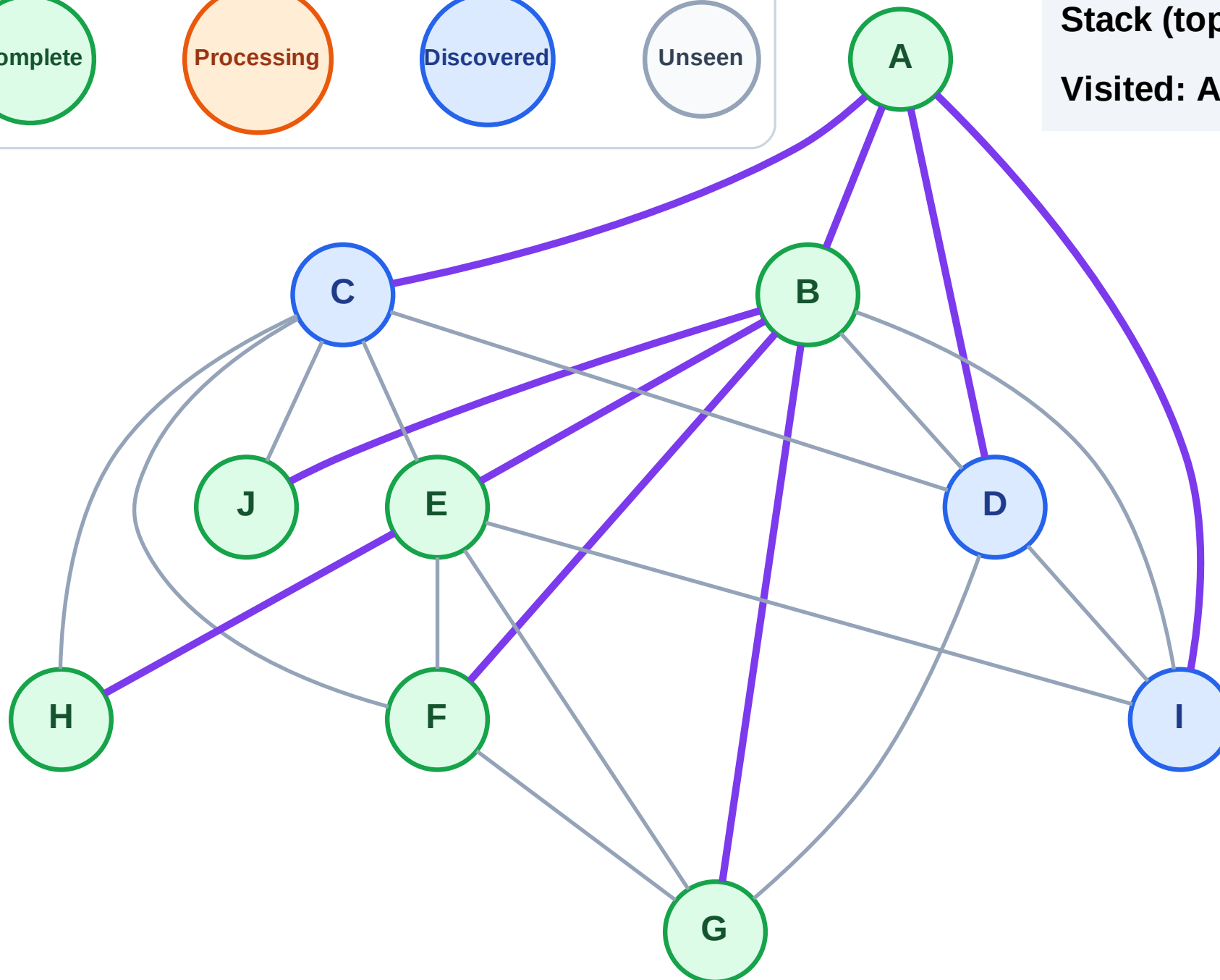
Step 15: No new neighbors from J

Legend



Stack (top at right): I | D | C

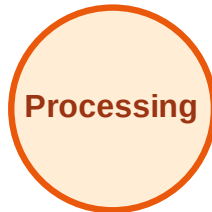
Visited: A, B, E, F, G, H, J



DFS Traversal

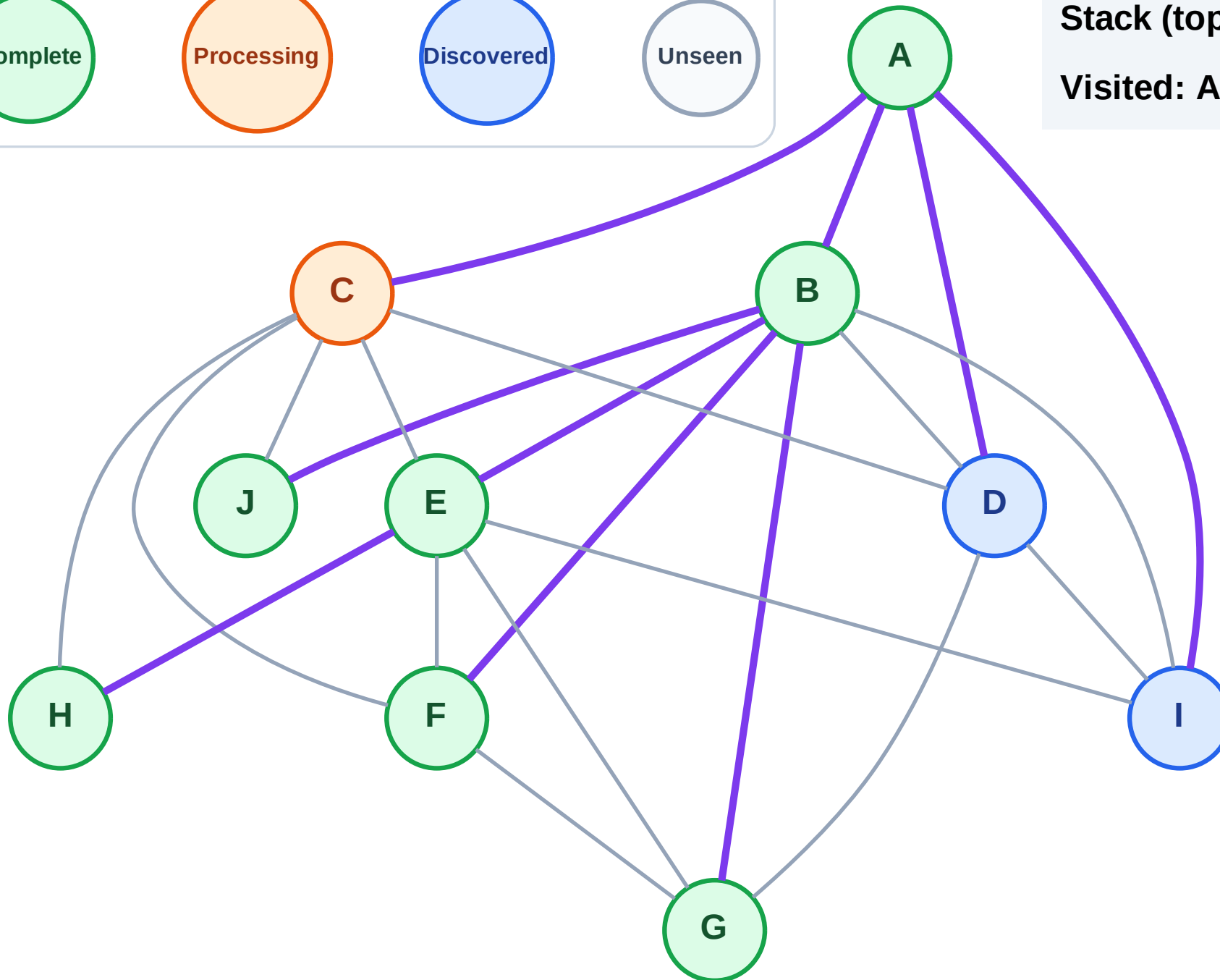
Step 16: Process node C

Legend



Stack (top at right): I | D

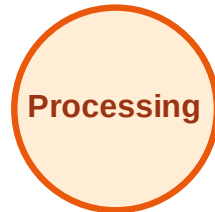
Visited: A, B, E, F, G, H, J



DFS Traversal

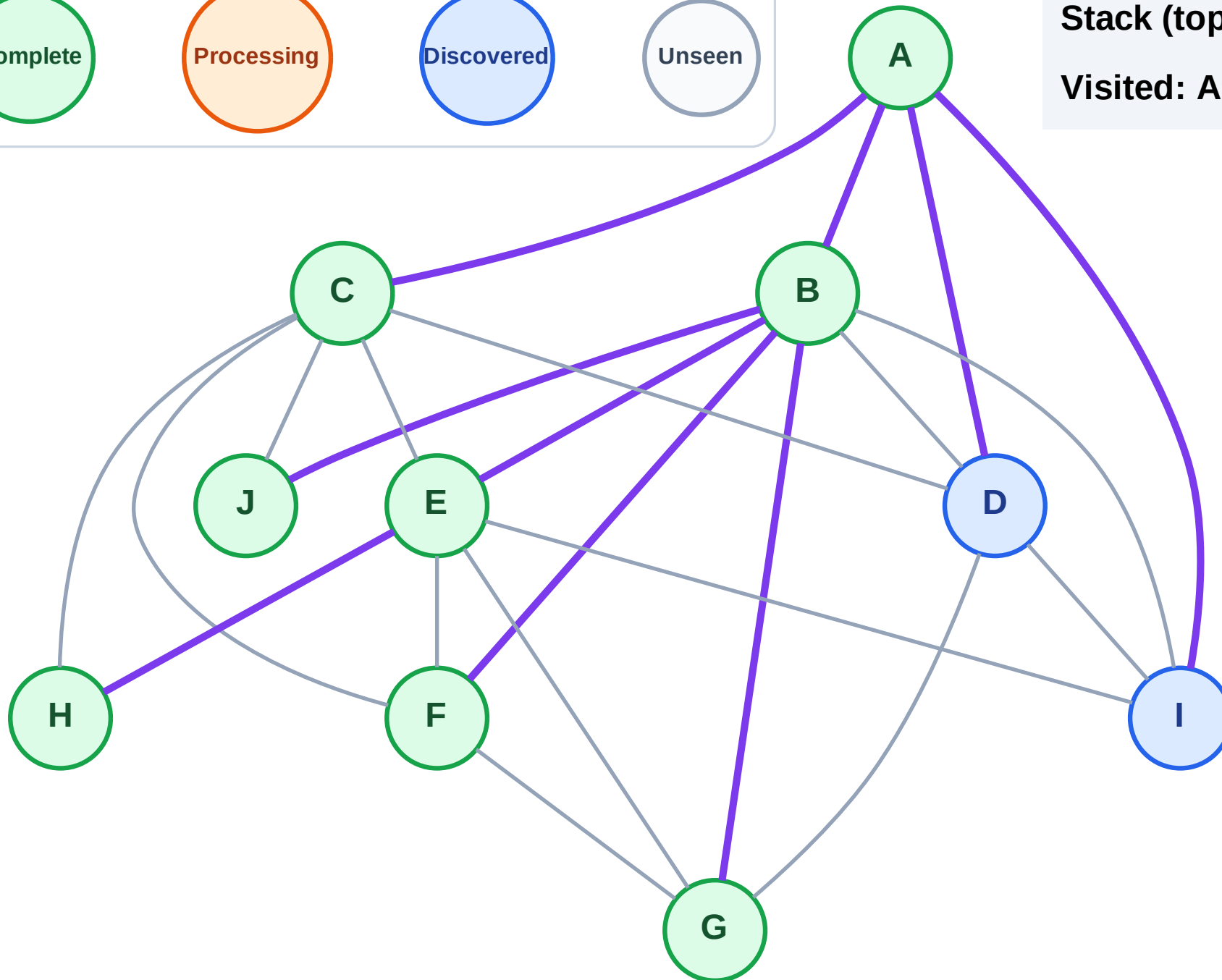
Step 17: No new neighbors from C

Legend



Stack (top at right): I | D

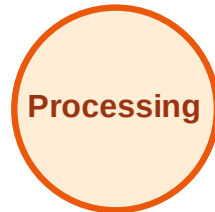
Visited: A, B, C, E, F, G, H, J



DFS Traversal

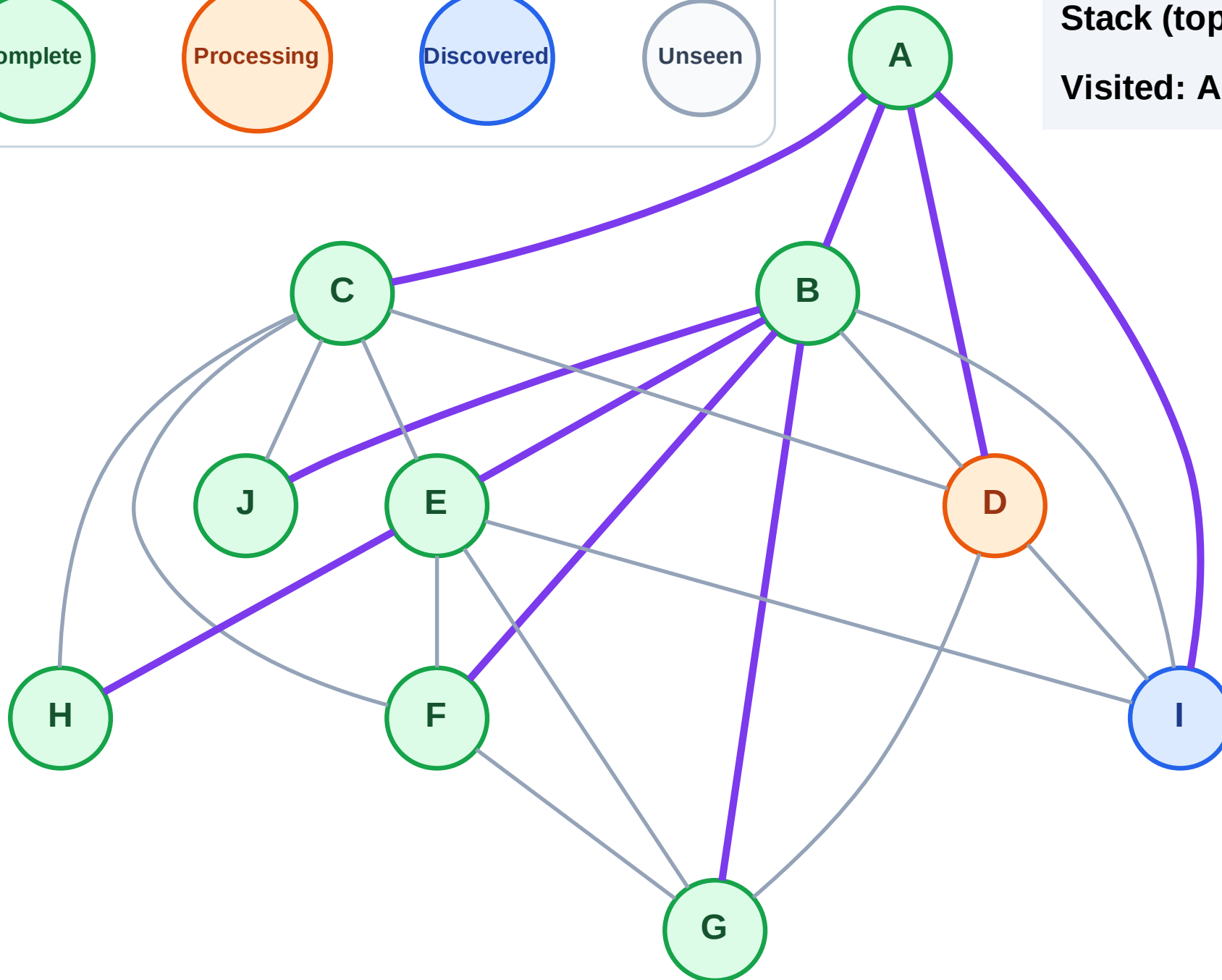
Step 18: Process node D

Legend



Stack (top at right): I

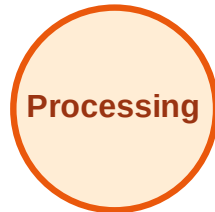
Visited: A, B, C, E, F, G, H, J



DFS Traversal

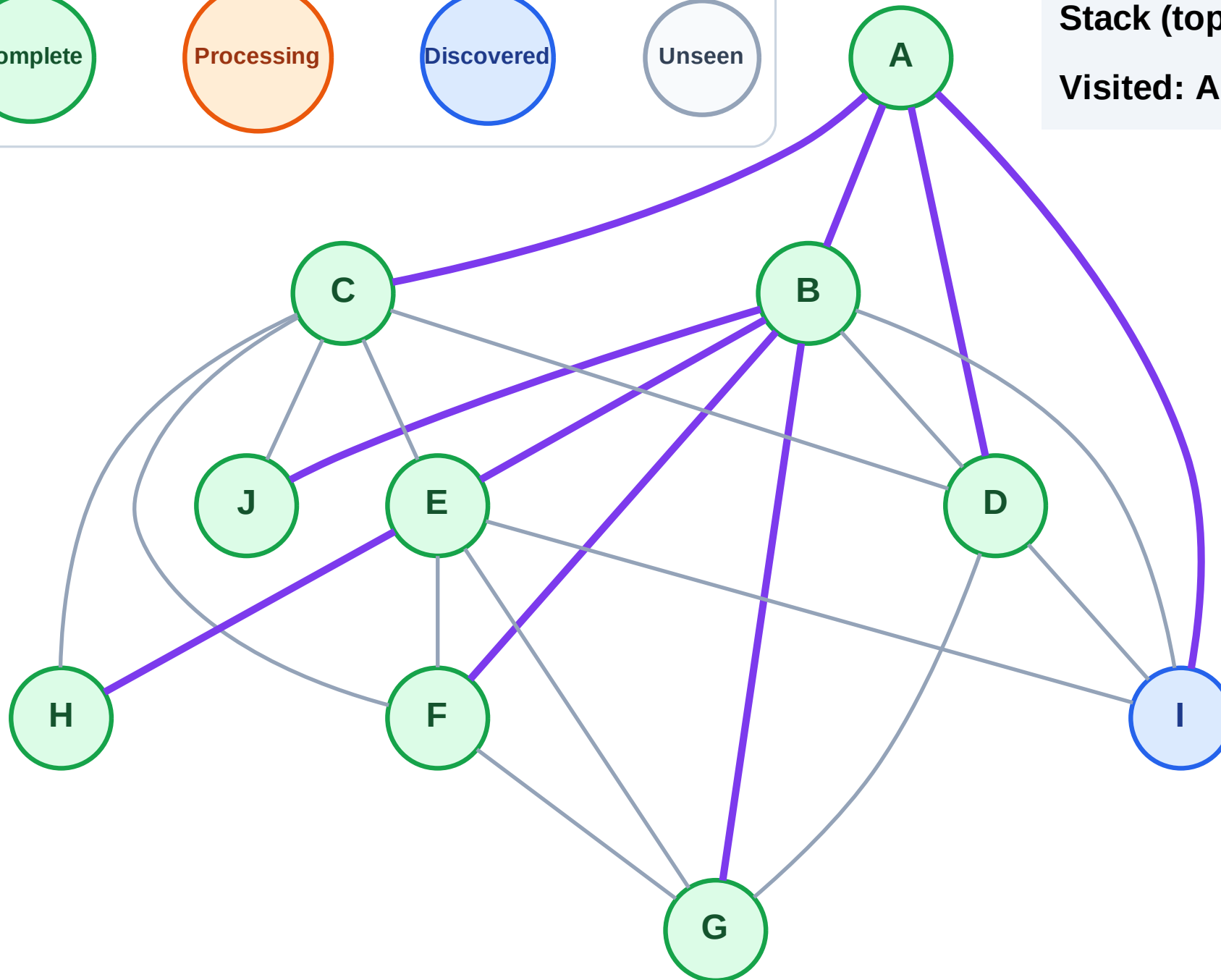
Step 19: No new neighbors from D

Legend



Stack (top at right): I

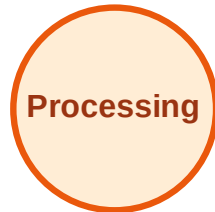
Visited: A, B, C, D, E, F, G, H, J



DFS Traversal

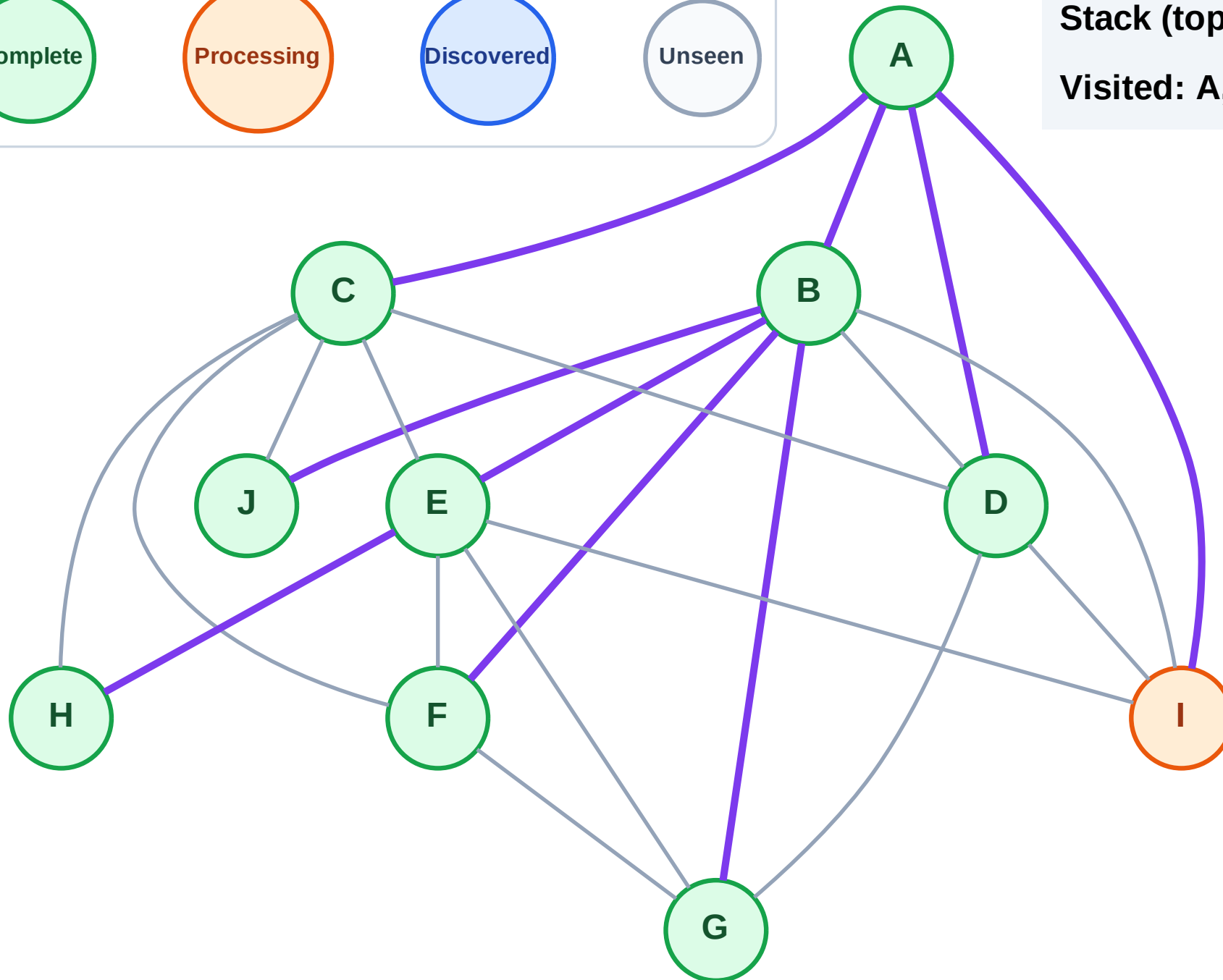
Step 20: Process node I

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, J



DFS Traversal

Step 21: No new neighbors from I

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

